Write What You Know: Topic Familiarity and Bias in Standardized Tests

Dalila Figueiredo [*] Lucas Finamor[†]

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Abstract

Standardized tests frame what they measure through topics: an essay about museums, a math problem about football. If familiarity with these topics differs across test takers and affects performance, tests may reward background rather than skill. By combining a survey of familiarity with real topics from standardized English tests and a randomized experiment among prospective test takers in Brazil, we find significant familiarity disparities across individuals by socioeconomic status, race, and minority status. We also find that familiarity causally affects test scores, including scores on criteria that measure language ability alone. Familiarity differences can account for over half of the socioeconomic score gap, making them an important determinant for test fairness.

1 Introduction

Access to higher education is a primary driver of upward mobility (Card, 1999; Goldin and Katz, 2008; Autor, 2014). Access often depends on performance in standardized tests, which aim to evaluate all test-takers by the same criteria. Societies use them to rank individuals, and their results serve as gatekeepers to education, jobs, and migration. Because standardized test performance can be so consequential, the fairness of these assessments is essential to ensure equal access to economic opportunities. When a test captures a candidate's background in addition to the skill it intends to measure, standardized evaluation may perpetuate inequality in access to opportunities.

[*]Max Planck Institute for Behavioral Economics, figueiredo@econ.mpg.de.
[†]Sao Paulo School of Economics, FGV lucas.finamor@fgv.br.

In this paper, we focus on an understudied source of test unfairness: topic familiarity. Many tests use everyday topics or contexts to frame questions, such as a math problem about the size of football fields or an essay prompt about museums. These topics are not meant to affect what is being measured, but they can. If some people are more familiar with a topic for reasons unrelated to ability, test scores will be biased in their favor. If familiarity differs systematically across social groups, scores reflect students' background, not just skill. To shed light on this issue, we ask three important questions: Are there topic-familiarity gaps across socioeconomic status, gender, or race? Does familiarity affect performance? And can simple interventions close these gaps, and for whom?

We answer these questions through an online experiment, conducted in Brazil, in which we exogenously vary individuals' topic familiarity. First, we elicit individuals' characteristics and their familiarity with 30 topics used in real exams. This allows us to document familiarity disparities across socioeconomic status, race, and gender. To test whether these familiarity gaps translate into performance gaps, we run an experiment. Study participants read a randomly assigned magazine article on a given topic, in their native language, which should increase their familiarity with that topic. Then, they write an essay on a predetermined topic. By comparing the performance of those individuals whose familiarity with the essay topic was exogenously increased with those in the control group, whose relevant familiarity remained unchanged, we estimate the causal effect of familiarity on writing scores. Finally, using an alternative treatment group, we contrast the performance effects of changing familiarity with the performance effect of letting individuals choose their preferred essay topic. With this design, we test which policy better closes the documented familiarity gaps.

A key challenge in identifying familiarity's effect on performance is that most tests measure topic familiarity or knowledge alongside other skills, making it hard to isolate familiarity's effect. For example, a widely discussed American SAT analogy item required knowing what a regatta is, turning a verbal reasoning question into a test of exposure to rowing culture (Kidder and Rosner, 2002)[1]. In Brazil's ENEM (High School National Exam), the bundling is intentional. The essay task rewards applying knowledge from several fields to develop the assigned theme, so topic knowledge of-

[1]The item became a standard exhibit in the public debate over SAT cultural bias (see Lemann (1999) for the history), and the College Board removed the analogy section altogether in 2005.

ficially counts toward the writing score, alongside language criteria. In tests like these, one cannot separate the effect of familiarity from other constructs being measured. To circumvent this problem, we use Standardized English Tests (SET), which aim to measure English ability rather than topic knowledge. This lets us identify any familiarity bias, since English skill should not depend on the test topic. We use real tasks from past tests and recruit participants interested in taking these exams, bringing the setting close to a real exam environment. Notably, the mechanism we identify and the measurement tool we build extend to other high-stakes standardized tests with topic-framed prompts, including the national university-entrance and school-leaving exams that dictate opportunities across countries. This setting gives us another important advantage. Because SET takers are non-native English speakers, we conduct the entire study, including the article, in participants' native language, ensuring the intervention only changes familiarity rather than teaching English-language skills.

A second challenge is that familiarity is not randomly assigned. Individuals who are more familiar with test topics also tend to have more education and higher income, and these characteristics predict performance on their own. A correlation between familiarity and scores could therefore reflect background characteristics, such as income, gender, or race, rather than any causal effect of familiarity. Randomly varying familiarity is difficult in real-world tests, and test topics are not chosen at random. Most exams also do not measure familiarity directly, making it hard to link familiarity effects to inequality. Therefore, an experiment is the ideal methodology for exogenously varying both topic familiarity and exposure.

Using the described setting and design, we find significant variation in familiarity across the 30 topics selected from past SETs. This is evidence that overall familiarity varies widely across writing tasks, which makes tests highly heterogeneous in this dimension. Then, we show that there are systematic familiarity gaps for Low-SES and non-White/non-Asian individuals. In particular, White and Asian participants are, on average, 0.07 standard deviations (s.d.) more familiar with the tests' topics than non-White/non-Asian participants, and High-SES participants are, on average, 0.13 s.d. more familiar than Low-SES participants, both significant at 1%. We find no gap for gender.

Having documented significant familiarity gaps with the topics in English Standardized tests, we next estimate whether these gaps matter for test performance. We

first confirm that our intervention successfully increased familiarity: reading an article on a given topic increases topic familiarity by 0.12 s.d. Note that this corresponds to the size of the SES familiarity gap described above. We then show that the intervention increases writing score by 0.13 s.d., equivalent to moving an individual from the 39th to the 48th percentile of TOEFL's worldwide writing proficiency distribution.[2] Finally, through an IV estimate, we show that a one standard deviation increase in familiarity increases the essay score by 1.1 s.d.

Next, using the estimates of topic-familiarity gap and topic-familiarity bias in SETs, we can decompose the socioeconomic and racial performance gaps and quantify the role of familiarity bias in them. We find that differences in topic familiarity alone can account for 0.12 s.d. of the score advantage of High-SES test takers, over half of the socioeconomic performance gap among control participants. On the racial gap, familiarity can account for about 0.09 s.d. of the score difference. Although imprecisely estimated, this magnitude is comparable to the entire racial gap we observe in the control group.

We also test a second intervention that is simple and cheap to implement: letting participants choose their essay topic. We test whether this alternative intervention leads to better performance than the current status quo (control group) and the article intervention. Participants rated their familiarity with six topics in the first part of the study, and in the second picked one of those to write about. This design tests whether choice increases performance and whether participants select their most familiar topic. We find that allowing individuals to choose their essay topic significantly increases their writing score. These individuals performed 0.27 s.d. better than the control group and 0.14 s.d. better than those who read the relevant article. Although we cannot directly test whether this effect is due to increased familiarity, the choices strongly validate our familiarity measure: 83% of individuals selected the topic they rated most familiar at baseline, and this topic is 0.6 s.d. more familiar than the non-chosen ones.

Finally, we test whether both interventions benefit some groups more than others. Overall, we rule out large differences across groups, but provide suggestive evidence that both policies might narrow the SES performance gap, while the free choice intervention may backfire on the racial gap. We conclude that although the interventions

[2]The 39th percentile is the position of our median control participant in the worldwide distribution.

do not benefit some groups more than others, both improve performance by increasing test-takers' familiarity.

Our paper adds to the growing literature on how test and assessment design shapes outcomes and can disadvantage certain groups (see Diris et al., forthcoming, for a review). Multiple studies show that penalizing wrong answers in multiple-choice tests creates a gender gap, as female test-takers skip uncertain questions more often than males (Baldiga, 2014; Riener and Wagner, 2017; Coffman and Klinowski, 2020; Iriberri and Rey-Biel, 2021; Saygin and Atwater, 2021). Closest to our work, Duquennois (2022) finds that math questions framed around money cause low-SES students to underperform, attributing this to an attention-capture mechanism where monetary framing triggers scarcity concerns. Banerjee et al. (2025) also show that maths skills acquired in real-world settings fail to transfer to the classroom and vice versa. While these two papers focus on monetary framing and SES, we examine a broader familiarity channel that can affect any group across topics, race, SES, and gender whenever test-takers differ in experience or knowledge.

Our study most closely relates to a contemporaneous work by Lee and Schaelling (2026), which provides large-scale, quasi-experimental evidence that content relatability affects test performance and contributes to racial test-score gaps. Using item-level data from Texas standardized reading exams and the American Time Use Survey, they build a race- and ethnicity-specific measure of passage relatability. Using a shift-share framework, they show that a one-standard-deviation increase in relatability raises test scores by about 0.05 standard deviations, accounting for roughly 4% of Black-white and Hispanic-white reading gaps in grades 3 to 8.

Relative to the existing literature, we extend this agenda in three important ways. First, our analysis measures familiarity and proficiency at the individual level, and therefore it does not rely solely on group-level differences. As a result, we can use it as a building block for higher-level analysis. In particular, it allows us (i) to explore heterogeneity within groups; and (ii) to apply the analysis to any relevant social dimension (gender, race, socioeconomic status, education, place of birth, etc.), including intersectionalities. Second, through the experimental design, we show that topic familiarity is not a fixed trait but is malleable. This is an important feature since it allows us to design and test policies that manipulate individuals' familiarity and compare them with alternative policies. Moreover, it allows us to test heterogeneous treatment

effects of such policies, shedding light on which of them are more effective at closing these systematic gaps. Third, we take this question to a uniquely relevant setting for fairness debates: standardized English tests (particularly writing tasks). These are designed to measure proficiency, not topic knowledge. This is in contrast with other exams where knowledge of the topic is also part of the evaluation. Our results show that performance gains from increased familiarity are uniform across all four writing evaluation criteria, including grammar and lexical resource, which should measure only language. Therefore, familiarity provides more than topic knowledge and spills over into language-only outcomes.

Finally, another key contribution of our paper is testing two practical policy interventions to reduce performance gaps. Dobrescu et al. (2021) evaluate a complementary, supply-side approach: a version of Australia's national basic-skills assessment redesigned to reflect the local context of a rural community. The contextualized test raised reading scores for both Indigenous and non-Indigenous students but lowered non-Indigenous students' numeracy performance, showing that redesigning test content is challenging and can produce unintended adverse effects. Our interventions operate instead on the test-taker side and leave the test almost unchanged. The first provides a short informational article on the essay topic in the test-taker's native language, similar to practices in Brazil's ENEM and Germany's Abitur. The second lets test-takers choose from a menu of topics, assuming they will select the one they know best. Both approaches are low-cost, maintain test validity, and could be widely adopted. We show that both interventions increase performance.

The remainder of the paper is organized as follows. Section 2 presents the conceptual framework and defines the estimates of interest. Section 3 details the experimental design and Section 4 the Empirical Strategy. In Section 5 we present the summary statistics and the results are detailed and analysed in Section 6. In Section 7 we introduce an alternative intervention which success in decreasing the famliarity-gap we compare with the experimental intervention. In Section 8 we present several robustness tests and Section 9 concludes the paper.

2 Conceptual Framework

Before detailing our experimental design, we present a simplified theoretical framework to clarify what we define as a biased test against certain social groups.

Let test taker i belong to a group $g \in \{H, L\}$. For example, g can be socioeconomic status: **H**igh or **L**ow SES. Their true English writing proficiency is a_i. When taking an SET, they write about topic k and obtain the score S_i. Individual i has some familiarity (background knowledge or personal experience) with topic k, which is given by f_{ik}. Without loss of generality, we assume $\mathbb{E}[f_{ik}] = 0$.

We define topic-familiarity ***gap*** as

$$\mathbb{E}[f_{ik}|i \in H] - \mathbb{E}[f_{ik}|i \in L] = \gamma. \tag{1}$$

If $\gamma = 0$, then individuals from groups L or H have the same average familiarity over the distribution of topics k that are covered in SET exams. If $\gamma > 0$, individuals from group H are, on average, more familiar with the topics.

On test performance, we define

$$S_{ik} = a_i + \beta f_{ik} + \varepsilon_{ik}, \tag{2}$$

that is, the score is given by the test-taker ability a_i, a potential effect of familiarity on topic k, and classic measurement error ε_{ik}. If $\beta = 0$, then topic-familiarity does not affect writing scores. If $\beta > 0$, individuals obtain higher scores when writing about more familiar topics.

We can now define topic-familiarity ***bias*** in SET as

$$\begin{aligned} \mu &= \mathbb{E}[S_{ik} - a_i \mid i \in H] - \mathbb{E}[S_{ik} - a_i \mid i \in L] \\ &= \beta \left(\mathbb{E}[f_{ik} \mid i \in H] - \mathbb{E}[f_{ik} \mid i \in L]\right) \\ &= \beta\gamma. \end{aligned} \tag{3}$$

If there is no topic familiarity gap across socioeconomic groups ($\gamma = 0$), then familiarity-bias is zero. If there are topic familiarity gaps, but familiarity does not influence test scores ($\beta = 0$), then familiarity-bias is zero. To assess whether standardized tests are biased against certain social groups, we need to test whether both $\gamma \neq 0$ and $\beta \neq 0$.

Importantly, we can also decompose the socioeconomic gap (or other group-gap) in test scores into

$$\begin{aligned}\mathbb{E}[S_{ik}|i \in H] - \mathbb{E}[S_{ik}|i \in L] &= \underbrace{(\mathbb{E}[a_i|i \in H] - \mathbb{E}[a_i|i \in L])}_{\text{ability gap } (\Delta a_i)} + \underbrace{\beta(\mathbb{E}[f_i|i \in H] - \mathbb{E}[f_i|i \in L])}_{\text{familiarity bias } (\mu)} \\ &= \Delta a_i + \mu \end{aligned} \tag{4}$$

Our experimental design allows us to test the following: (i) whether $\gamma \neq 0$; (ii) whether $\beta \neq 0$; and (iii) which interventions may address closing the bias, by either manipulating f_{ik} or changing k, and whether they might work better for some groups than others. It also allows us to quantify the weight of the familiarity gap on test scores for the different social groups.

It is important to highlight that our definition of ***bias*** μ reflects a systematic bias across groups that depends on $\gamma \neq 0$. This differs from a statistical-bias notion, which only requires $\mathbb{E}[S_i|a_i] \neq a_i$. In this setting, SET can still be biased in a statistical sense if $\beta \neq 0$ and $\mathbb{E}[f_{ik} \mid a_i] \neq 0$, but have no systematic bias against any group if $\gamma = 0$. Additionally, even when $\mathbb{E}[f_{ik} \mid a_i] = 0$ and $\gamma = 0$, $\beta \neq 0$ alone implies that realized scores deviate from ability by βf_{ik}, so individual test takers are still mismeasured. Individuals may get higher or lower scores depending on the luck of their topic draw. In Section 8, we provide evidence of this last channel.

3 Experimental Design

3.1 Standardized English Tests

We design an experiment around the writing task of Standardized English Tests (SET), such as IELTS or TOEFL. These tests usually have four components: Listening, Speaking, Writing, and Reading. We focus on those tasks that consist of a statement to which the test-taker has to agree or disagree, and on providing arguments in favor of or against the statement and their point of view.[3] We collected a set of statements from publicly available past exams online and randomly selected 30. These statements, or *topics*, as we call them throughout the paper, are used to document familiarity dis-

[3]For instance, in the IELTS exam, this corresponds to the second writing task.

parities across individuals from different socioeconomic statuses, genders, and races.

From these 30 topics, we selected 6 to study how familiarity affects performance. Each participant must write an essay on one of these 6 topics. Running the experiment with 6 different topics yields more generalizable results for the test itself. We refer to these 6 topics as the *experimental-topics*. Table A1 shows all the selected topics, starting with the six that are used in the writing task. The task follows the format commonly used in SETs, as shown in the example in Box 3.1

Box 3.1

You should spend about 40 minutes on this question.
Write about the following topic:

Team activities can teach more skills for life than other activities which are played alone. To what extent do you agree or disagree?

Give reasons for your answer and include any relevant examples from your own knowledge or experience.

Write at least 250 words.

3.2 Experimental Design

The study is divided into two parts (I and II).

Part I serves two main purposes: to document familiarity disparities across socioeconomic status, gender, and race for 30 topics from past real examinations; and to measure baseline familiarity, background knowledge, and other important predictors of writing performance. Therefore, the first survey collects (i) demographic characteristics, (ii) familiarity with 8 topics, (iii) a general knowledge test, (iv) an English diagnostic test, and (v) a critical thinking scale.

In Part I, we ask each participant to rate their familiarity with 8 topics: (i) 2 of these topics belong to the 6 *experimental-topics*; (ii) 4 of these were randomly selected from the remaining 24; and (iii) the remaining 2 are anchoring topics that are used to bound our familiarity measure, as described in Section 4.1. Importantly, the general knowledge quiz also covers topics from the 6 *experimental-topics* and is used as a validation of our measure of familiarity.

Individuals who complete Part I receive a link to complete Part II, 2 to 3 days after

completing Part I. In the second survey, they are asked to (i) read a magazine-like article, in their native language, concerning one of the *experimental-topics*, randomly selected, and answer comprehension questions; (ii) rate their familiarity with the two topics of the relevant group; (iii) write an essay on a randomly chosen topic within the topic group; and (iv) answer further questions on other personal characteristics. Both surveys are in the participants' native language. The only parts of the survey that are in English are the English diagnostic test and the essay task. Part I and II of the survey include attention checks.

The *experimental-topics* are always considered in pairs. For those individuals randomized into pair A, the experiment covers topics 1 and 2. This means they rate their familiarity with topics 1 and 2, answer general knowledge questions on these topics, read an article on one of them, and write about either topic 1 or 2. Similarly, those randomized to pair B interact only with topics 3 and 4. Finally, pair C participants interact only with topics 5 and 6.

In addition to randomizing topic exposure, our experimental design further randomly manipulates whether an individual reads the article on the topic they are asked to write the essay on. We randomly allocate each participant into one of 5 treatment arms: T, C1, C2, C3, and A. Each participant has a 37.5% probability of being in treatment arms T and C1, a 15% probability of being in treatment A, and a 5% probability of being in treatments C2 and C3.

The main analysis relies on arms T, C1, and A. Individuals allocated to treatment arm T (the treatment group) read an article on the same topic as the writing task. Individuals in the first control group, C1, differ from T only in that they read an article on a different topic from their writing task. Comparing individuals in treatment arms T and C1 who wrote an essay on topic k allows us to identify the causal effect of familiarity with topic k on the essay's writing score.

Individuals in treatment group A, the alternative control group, are allowed to choose which of the 6 *experimental-topics* they write about. Importantly, in part 1, participants randomly allocated to this group answer familiarity questions on the 6 aforementioned topics and the two anchor topics. They do not read any articles in survey 2 or answer any further familiarity questions. This group tests an alternative policy intervention that is easy to implement and replicate across different tests. We want to test whether individuals perform better when they can choose their preferred topic, how

this alternative compares to providing topic information before writing (T), and which intervention best closes the familiarity-induced performance gap across individuals from different SES backgrounds, genders, and races.

Treatment arms C2 and C3 are very similar to C1, in that the article they read before writing the essay is on a different topic than the writing task. Additionally, C3 does not answer familiarity questions after reading the article, but instead only after writing the essay. Alternatively, C2 reads a second article on the topic of the essay **after** having to write the essay and then answers familiarity questions. We use these two groups for some robustness tests.

Table 1 below shows an example in which everyone is allocated to topic pair A and is randomly chosen to write essays on topic 1 (rather than topic 2). We would have similar tables for all pairs and topics.

Table 1: Groups - Example: Those in Pair A That Write an Essay on Topic 1 (T1)

Group	(notation)	T	C1	C2	C3	A
Share		37.5%	37.5%	5%	5%	15%
Survey Part I						
Demographics		✓	✓	✓	✓	✓
Familiarity	(f^{s1})	T1/T2	T1/T2	T1/T2	T1/T2	6 topics
Familiarity Anchoring	(f^{anc})	✓	✓	✓	✓	✓
Additional Familiarity	(f^{add})	✓	✓	✓	✓	-
General Knowledge		✓	✓	✓	✓	✓
English Diagnostic Test		✓	✓	✓	✓	✓
Critical Thinking Scale		✓	✓	✓	✓	✓
Survey Part II						
Article		T1	T2	T2	T2	-
Familiarity	(f^{s2})	T1/T2	T1/T2	-	-	-
Essay	(y)	T1	T1	T1	T1	Choice
Additional Article		-	-	T1	-	-
Familiarity II	(f^{s2a})	-	-	T1/T2	T1/T2	-

Note: This table shows the components of each survey part by treatment arm, for the example of participants in pair A whose essay topic is T1. Checkmarks indicate components administered identically to all arms. Topic entries indicate which topics the component covers, and Share gives the allocation probability of each arm.

3.3 Recruitment and Sample

Our population of interest is Brazilian individuals between 18 and 40 who are potential Standardized English Test takers. We target this population through social media ads, promoting the study as an opportunity for those wanting to study or work abroad to practice and test their English Writing proficiency. Brazil is an ideal context for this study for two main reasons. First, since the main goal of this paper is to document socioeconomic, racial, and gender disparities, we need a country in which the population of interest is diverse along these lines. The country's size and the policies implemented over the last decades that incentivize Low-SES and Black students to access university mean there are many students interested in accessing education and labor market opportunities nationally and internally who must take SET as proof of English proficiency. Second, Brazilian Portuguese, the country's native language, is sufficiently distant from English that we can conduct the study and manipulate participants' familiarity without compromising their English proficiency. This helps ensure clean identification.

Participants receive a fixed participation fee, depending on completion and answer quality, if they complete the study (parts I and II). To incentivize careful reading, we reward individuals for correctly answering the comprehension questions after reading the article. We randomly select 1 out of 10 participants and pay them a bonus if they get at least 80% of the questions right. To further encourage them to exert effort when writing the essay, we pay a large prize for the best 10% of essays. Finally, we offer them the opportunity to practice in a setting very close to the real test and receive feedback on their essay, which can help participants assess their English writing level.

To ensure question quality, we impose several criteria. We have two pre-registered sets of criteria. The conditions in Criteria I are: (i) to complete at least 50% of survey I; (ii) to answer the familiarity questions of survey I in strictly more than 10 seconds; (iii) to answer the anchoring familiarity questions in a coherent way; (iv) to answer correctly at least two (out of the three) attention check questions. Participants have access to Part II of the study only if they meet all these conditions. Then, we further impose the following Criteria II: (v) to complete both surveys; (vi) to answer the familiarity questions in both surveys in strictly more than 10 seconds; (vii) to stay on the article page for at least 90 seconds; (viii) to answer correctly one (out of 2) attention check questions. We include all participants who satisfy Criterion I in the final sample for

the analysis documenting disparities across groups. For the experimental analysis, we consider only individuals who satisfy both sets of criteria.

Trivially, we have removed all observations flagged as **bots** and those from individuals who used AI to write their essays, as we are interested in human behavior.[4] Importantly, treatment allocation does not predict the usage of AI for writing (see Table A4). Nevertheless, for robustness, we present results that include those observations in Appendix A.3.

Of the 9,195 individuals who reached half of Survey I, 3,982 met the criteria in Set I. Of those, 952 participants met the criteria in Set II, and 944 wrote a valid essay, defined as an essay written in English and on topic. Of these, we detected 166 essays written with the support of AI. We excluded them from the sample, leaving a final sample of 778 observations.

Table 2: Sample

	N
Reached at least 50% of Survey I	9,195
Satisfy criteria I	3,982
Satisfy criteria II	952
Valid essay writers satisfying criteria II	944
Valid essay writers satisfying criteria II, human-written	778

Note: This table shows the progression of participants through the quality criteria. Criteria I and II are described in Section 3. A valid essay is written in English and on topic. The final sample keeps only the essays classified as fully human-written by *Pangram*.

4 Empirical Strategy

4.1 Outcomes

In this paper, we focus on two main outcomes: Familiarity and Writing Proficiency.

Familiarity We define familiarity with a topic as the set of experiences and knowledge an individual has regarding that topic, as well as the confidence and ability to

[4]For bots, we rely on classification implemented by *Qualtrics*. To identify AI usage, we assess all essays on *Pangram* and discard those that are not classified as 100% humanly written.

discuss it in different contexts.

We have developed a self-reported familiarity measure based on Likert-scale responses to 6 statements. In the first 5, we ask individuals whether each of the following sentences is definitely false, somewhat false, neither true nor false, somewhat true, or definitely true.

(f_1) I have knowledge about this topic because I have studied or read about it.

(f_2) I have knowledge about this topic because I have had personal experience with the issues it addresses.

(f_3) I have thought about and/or discussed this topic before with family, friends or at school.

(f_4) I am capable of discussing this topic informally with friends or family.

(f_5) I am capable of discussing this topic in a formal context and providing arguments for or against the statement.

The last statement (f_g) asks individuals how familiar they are with the topic of the statement above. There are 5 possible answers:

1. Nothing familiar. I wouldn't be able to discuss the matter.
2. Unfamiliar. I would have difficulty discussing the matter.
3. Somewhat familiar. I could discuss the matter but with some difficulty.
4. Familiar. I would be capable of discussing the matter.
5. Very familiar. I would be extremely capable of discussing the topic.

Questions f_1, f_2, and f_3 assess what individuals believe they know about the topic. Questions f_4, f_5, and f_g elicit how familiar individuals feel about a given topic, and how they would be able to discuss it. To construct a measure of familiarity, we use f_1, f_4, f_5, and f_g, since we believe this is what we can manipulate in this study.[5]

[5]In the pre-analysis plan, we pre-registered a familiarity measure based on f_4, f_5, and f_g. However, our treatment (the article) operates through the knowledge dimension captured in f_1, since participants acquire knowledge about the topic precisely be reading about it. Hence, including this dimen-

$$\tilde{f} = \frac{1}{4}(f_1 + f_4 + f_5 + f_g)$$

Since this measure of familiarity is subjective, we also include two anchoring topics. These aim to improve the measure's accuracy, comparability, and interpretability across participants. One topic should be very familiar to most of our subject pool: 'It is more enjoyable to do outdoor activities when it is not raining than when it is raining.'; and the second should be very unfamiliar to our subject pool, since it is nonsensical: 'Countries should invest more in the dynamic exploration of extra-marginal resources.'. We use these two questions to create an upper, $\tilde{f}^u$, and a lower, $\tilde{f}_l$, bound on familiarity.

Our final familiarity measure rescales the familiarity score $\tilde{f}$ using the answers to the very-familiar and not-familiar anchoring questions.

$$f = \frac{\tilde{f} - \tilde{f}_l}{\tilde{f}^u - \tilde{f}_l}$$

In case $\tilde{f}$ is outside the range $[\tilde{f}_l, \tilde{f}^u]$, we truncate its value within the range limits.[6] We standardize this measure to have a mean of 0 and a standard deviation of 1, across topics (for the first research question) and within topics (for the second research question). The first three questions help us understand why familiarity gaps exist and test how our treatment affects familiarity. [7]

Section 7 provides a behavioral validation of this measure. When participants can choose their essay topic, 83% choose the topic they rated most familiar days earlier, in an incentivized decision.

sion in the familiarity measure captures fully the changes the treatment had on familiarity. Furthermore, the incentivized comprehension questions after the article confirm that reading the article translated into acquired knowledge, so the change in f_1 reflects genuine knowledge acquisition rather than a mechanical response to having read a text. Throughout the paper, our familiarity measure therefore includes f_1. For transparency, we report all results using the pre-specified measure in the Appendix, and we refer to them where relevant when discussing the results.

[6] This is equivalent to truncating the final variable in the 0–1 range.

[7] Figure A1 shows the distribution of familiarity with the two anchoring topics. The anchors behave as intended. For 98 percent of participants, the familiar anchor is rated strictly above the nonsensical one, with an average gap of 2.2 points on the 1 to 5 scale, so the real topics fall inside almost everyone's personal range. At the same time, neither distribution is degenerate. The median ratings are 1.5 and 4.25 rather than 1 and 5, and participants differ in how they use the extremes of the scale. This variation is what gives the anchoring correction its content. The correction also absorbs response styles. One tenth of participants rate the nonsensical topic 3.5 or higher, a tendency to report high familiarity regardless of content that would inflate raw self-reports.

Writing proficiency We obtain our performance measure by prompting a large language model (LLM) to follow the standard guidelines from SETs. This generates 5 scores: Task Response, Cohesion and Coherence, Lexical Resources, Grammar, and Overall. We use the "Overall" measure as our primary outcome, standardized to have a mean of 0 and a standard deviation of 1 in the control group. We refer to this variable as y. Importantly, we feed the LLM only the essay. In the pilot, a retired official SET examiner scored the essays. The correlation between the LLM and the grader's scores was 0.66. The appendix includes the prompt used to guide the LLM.

Cheating We time how long individuals spend off the survey page while writing the essay. Participants are informed that if they leave the page, their essay is deemed invalid, and they will not receive a grade or the participation fee. We also warned individuals that their essays must be original, and the examiner is trained to detect plagiarism. These measures are intended to prevent cheating. We flag cheating if individuals left the essay page for more than 30 seconds, had their essay detected as plagiarized, or used AI assistance.[8]

4.2 Estimation

4.2.1 Disparities across groups

To compute the disparities in familiarity across groups, we estimate this main specification using a pooled OLS estimator:

$$f_{ik} = \gamma G_i + \eta_k + \delta X_i + \nu_{ik} \tag{5}$$

Where f_{ik} is the familiarity for individual i and topic k, G_i is a dummy vector indicating whether individual i belongs to group G. X_i is a vector of variables at the individual level, including age. In this step, we analyze the following groups: gender, race-ethnicity, education, parental education, and self-perceived minority status. The coefficient of interest is γ, which corresponds to the parameter described in Equation (1). We clustered the standard errors at the individual level. This analysis considers all individuals who passed our set of criteria I.

[8]Interestingly, and as a validation of both flags, we see that essays flagged as using AI assistance are 581% more likely to leave the page.

4.2.2 Causal Effect of Familiarity on Performance

First Stage: To estimate the causal effect of reading an article on a given topic on familiarity with that same topic, we estimate the following regression:

$$f_{ikt} = \alpha_1 \text{Info}_{ik} + \alpha_2 \text{Post}_{it} + \alpha_3 \text{Info}_{ik} \times \text{Post}_{it} + \eta_i + \nu_k + e_{ikt} \tag{6}$$

Where Info_{ik} is a binary indicator of whether individual i was randomly allocated to read the article on topic k. Post_{it} is an indicator for answering the familiarity question on survey II (after reading one article). We clustered the standard errors at the individual level.

Note that individuals respond to the familiarity questions on the two topics of the same pair, both before and after reading the randomly chosen article. Therefore, we can use within-individual variation to identify the effect of interest while controlling for individual and time-specific changes.

Reduced Form: To estimate the causal effect of reading the article on the same topic as the essay on writing performance, we estimate the following:

$$y_i = \pi \text{Treat}_i + \delta Z_i + u_i \tag{7}$$

Where y_i is the standardized essay score for individual i, Treat_i is a binary indicator of whether individual i was randomly allocated to write the essay on the same topic they read the article on. Z_i is a vector of variables at the individual level including age, gender, race, SES, education level, and scores on an English diagnostic test and a critical thinking scale. The coefficient of interest is π. We clustered the standard errors at the individual level.

Instrumental Variable: To estimate the effect of familiarity on writing performance, parameter β in Equation (2), we use the two estimates above and compute the 'instrumental variables estimator' as,

$$\hat{\beta}_{IV} = \frac{\hat{\pi}}{\hat{\alpha}_3} \tag{8}$$

To account for uncertainty in the two stages, we compute standard errors using a bootstrap procedure that accounts for individual-level clustering. Because our IV

estimator is the ratio of the reduced form to the first stage, conventional inference is unreliable, since with a moderately strong first stage, the sampling distribution of a ratio is heavily skewed and fat-tailed, and neither the bootstrap standard error nor a normal confidence interval is meaningful. In Figure A2, one can see the bootstrap distribution of the estimator and observe that it is non-normal. We therefore conduct inference by inverting an Anderson-Rubin test. For each candidate value β_0 we test the null $\beta_{IV} = \beta_0$, which is equivalent to testing that the linear combination $\hat{\beta}_{RF} - \beta_0 \hat{\alpha}_{FS}$ equals zero. Unlike the ratio, this combination is well behaved regardless of the strength of the first stage, and we compute its standard error from the same Bayesian bootstrap draws used throughout, which resample individuals and re-estimate both stages jointly, preserving their correlation. The 95 percent confidence set collects all values of β_0 that are not rejected at the 5 percent level. This procedure is robust to weak instruments, and we report the resulting interval in brackets in the results table.

We use this approach instead of a two-stage least squares (2SLS) strategy to leverage within-individual estimates of changes in familiarity following the informational treatment. This design accounts for individual scale usage and baseline familiarity, common time trends, and uses the paired non-article topic as a within-person placebo, delivering a richer and more transparent first stage than a cross-sectional 2SLS would allow.

The sample for the first stage, reduced form, and IV analyses comprises all individuals who met Criteria II and were randomized to groups T and C1.

5 Summary Statistics

To assess familiarity bias across topics, we use all participants who met Criteria I. These are 47% High-SES, which we define as having a mother with a College degree or above, 32% males[9], 60% white or Asian, 53% declare themselves as part of a minority group, are on average 25.7 years old, and 80% have a college degree (Table 3). To study whether familiarity affects performance, we impose Criteria II (and a human-written essay). In this final sample, 53% of participants are High-SES, 36% are male, 65% are White or Asian, and 50% self-declare as belonging to a minority group. They are, on

[9]The variable male is 1 if the individual declared themselves as male, and 0 if they are female, non-binary, or other.

average, 25.7 years old and 85% have a college degree (see Table 3). This corresponds to our target population. We aimed to promote the study to individuals likely to take the test while maintaining heterogeneity in socioeconomic status, race, and gender. Our successful targeting is further supported by self-declared data on their relationship with the test: 13% have already prepared for an English Standardized Test, 42% have already taken one, and 81% plan to. Considering the control group only, the average writing score is 5.7, which is slightly below, but close to the Brazilian average of 5.9[10].

Table 3: Descriptive Statistics

	Criteria I sample			Final sample		
	N	Mean	Std Dev	N	Mean	Std Dev
High-SES	3,943	0.465	0.499	772	0.534	0.499
Male	3,954	0.319	0.466	775	0.357	0.480
White/Asian	3,918	0.597	0.491	765	0.656	0.475
Self-Minority	3,980	0.527	0.499	777	0.488	0.500
Age	3,982	25.719	5.679	778	25.728	5.685
College Degree or above	3,982	0.799	0.401	778	0.851	0.356
Already prepared for a test	-	-	-	777	0.126	0.332
Already took a test	-	-	-	777	0.418	0.494
Is planning to take a test	-	-	-	777	0.804	0.397
Overall writing score	-	-	-	359	5.692	0.998

Note: This table shows individual characteristics for the sample of individuals who satisfied Criteria I (first three columns) and for the final sample of individuals who satisfied Criteria I and II and have a human-written essay (last three columns). The survey II variables are only available for the final sample. The Overall writing score is only displayed for individuals randomly assigned to the control groups.

We randomly assigned participants across the five treatment arms, according to the pre-specified proportions (Table A2). Note that Part I is identical for all individuals, regardless of which treatment arm they were assigned to. The differences across treatment arms appear only in survey II. Therefore, to measure attrition, we look at the difference between those individuals who have passed criteria I and those who have passed criteria II and had a human-written essay.[11] We have an average retention rate

[10] This average score includes 33% of the grade of the Writing Task 1, whereas in our study we only consider Task 2

[11] 93% of individuals who passed criteria I have declared their willingness to participate in survey II.

of 20%, which does not differ across treatment arms. Importantly, all characteristics of interest are also balanced across treatment arms (Table A3).

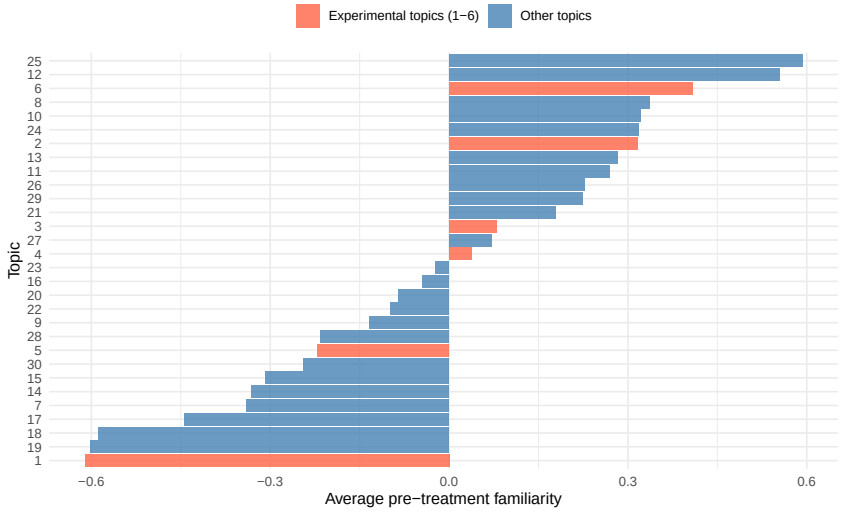
6 Results

6.1 Familiarity Disparities

We begin by documenting familiarity disparities across social groups defined by socioeconomic status, gender, and race.

First, note that the 30 topics selected for the first part of the study show a wide range of baseline familiarity. Even without looking at differences across social groups, this indicates that the tests vary widely in overall familiarity, depending on the topic chosen (See Figure 1). This variation matters because familiarity predicts performance. Among control participants, one standard deviation higher familiarity with the essay topic is associated with 0.07 standard deviations higher scores. This correlation already suggests that the topic a candidate happens to draw can move their score. In the next section, we show that this channel establishes a causal effect and we quantify it.

Figure 1: Familiarity Distribution across Topics



Note: The figure shows the average pre-treatment familiarity for each of the 30 topics, using the anchoring-adjusted measure standardized across all topics. The sample includes participants satisfying quality criteria I. Topics are ordered by average familiarity. The six experimental topics, those used in the essay task, are shown in red, and the remaining topics in blue. Topic numbers refer to the statements listed in Table A1.

Even if familiarity affects performance, this heterogeneity need not be a systematic

problem for test fairness if it varies randomly across individuals. Because it is hard to find completely neutral topics for everyone, individuals may be lucky or unlucky when sitting a test and writing about a familiar or unfamiliar topic.

However, this heterogeneity is a source of bias if it falls disproportionately on some groups more than others. Table 4 shows that this is the case for some of the analyzed groups. White and Asian individuals in Brazil are 0.07 standard deviations (s.d.) more familiar with the exam topics than non-White individuals. Similarly, High-SES participants are 0.13 s.d. more familiar with the covered topics than Low-SES ones. We find no evidence of a gender-based familiarity gap. The point estimate is extremely small and not statistically significant. These patterns hold even when we account for all characteristics simultaneously. Interestingly, familiarity is slightly larger (0.03 s.d.) for those who declare themselves as belonging to a minority than for those who do not, significant at 10%. Furthermore, among people of the same race and parental education, those who self-identify as minorities report higher familiarity of 0.07 s.d. In this study, minority is a self-declared, umbrella term. One can define themselves as belonging to a minority group depending on their race, income, gender, sexual orientation, religion, disability, or region or country of origin. Once race and parental education are held fixed, the remaining variation in the indicator reflects non-racial and income dimensions, plus the propensity to self-identify. Minority identifiers are more likely to be non-white, have lower parental education, and lower scores on the English diagnostic test, characteristics that predict lower familiarity. Conversely, they are more likely to be enrolled in university and score higher on the critical thinking scale, consistent with a positive correlation once controlling for race, gender, and SES.

Our familiarity measure allows us to test the origin of these gaps. Figure A3 decomposes them across the six items of the familiarity measure.[12] The SES gap is concentrated in the exposure items. High-SES participants report more contact with the topics through reading, thinking, and everyday conversation, with gaps around 0.11 s.d., about twice the gaps in personal experience or in the ability to discuss a topic. The racial gaps show a similar profile. The gender comparison confirms the absence of a familiarity gap: male participants report no advantage in any exposure item and differ only in the confidence to discuss topics formally. Finally, self-declared minorities

[12]Recall that each item is anchored by the respondent's own answers to the two anchoring topics and standardized, so the coefficients are comparable across items.

report higher familiarity uniformly across all six items. This flat profile is consistent with the propensity to self-identify discussed above, rather than with knowledge of specific topics. Taken together, the decomposition indicates that familiarity gaps reflect different exposure to the test's topics in daily life, whether in school or in their social circle.

If we consider only the 6 topics used in Part II of the study, we find similar patterns (see Panel B of Table 4). We find no gender gap in familiarity. White and Asian individuals are 0.09 s.d. more familiar than non-White/Asian individuals, and High-SES individuals are 0.11 s.d. more familiar than Low-SES individuals. The only pattern that does not hold in the last case is for the minority identifiers. For these six topics, individuals who self-declared as a minority show no familiarity gap. Since this group is less stable, we will focus the analysis on race and SES. Nevertheless, future research on their composition and intersection with other demographic dimensions would be worthwhile.

The consistency of the familiarity-gap findings between the 20 and the 6 topics is reassuring for Part II. We selected topics for the experiment across the familiarity distribution of the 30 topics and show similar disparities across the groups of interest (See Figure A4 for an illustration of how these topics rank in terms of familiarity for each group compared to the full pool).

Table 4: Familiarity Biases

Dependent Variable:	Familiarity (std)				
Model:	(1)	(2)	(3)	(4)	(5)
Panel A: All 30 topics					
Age	-0.0021	-0.0020	-0.0011	-0.0017	-0.0013
	(0.0017)	(0.0017)	(0.0017)	(0.0017)	(0.0017)
Male	0.0056				0.0210
	(0.0213)				(0.0216)
White/Asian		0.0741***			0.0735***
		(0.0200)			(0.0210)
High-SES			0.1280***		0.1293***
			(0.0192)		(0.0198)
Self-Minority				0.0341*	0.0720***
				(0.0193)	(0.0201)
Observations	23,201	22,998	23,140	23,356	22,602
FE: topic	Yes	Yes	Yes	Yes	Yes
Panel B: 6 experimental topics					
Age	0.0025	0.0026	0.0034	0.0027	0.0030
	(0.0023)	(0.0023)	(0.0023)	(0.0023)	(0.0023)
Male	0.0226				0.0377
	(0.0290)				(0.0295)
White/Asian		0.0874***			0.0815***
		(0.0269)			(0.0285)
High-SES			0.1105***		0.1100***
			(0.0259)		(0.0268)
Self-Minority				-0.0016	0.0307
				(0.0259)	(0.0274)
Observations	10,097	9,997	10,077	10,161	9,817
FE: topic	Yes	Yes	Yes	Yes	Yes

Note: This table shows the correlation between belonging to a certain group and baseline familiarity. Panel A aggregates the 30 test topics, where each participant declared their familiarity with 6 of the 30-topic pool, plus two anchors. Panel B aggregates the 6 test topics used in Part II, where each participant declared their familiarity with 2 of the 6-topic pool, plus two anchors. Both panels include all participants who complied with quality Criteria I. Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

In summary, we find a significant and robust familiarity gap for Low-SES and Non-White individuals. Therefore, for most of the paper, we will focus on these two dimensions. These groups seem most vulnerable to test design if familiarity indeed affects performance. In the next section, we study the causal effect of familiarity on test performance.

An advantage of having an individual, topic-specific familiarity measure is that it also lets us study within-group familiarity heterogeneity. Figure A5 plots the distributions of individual familiarity within each group: SES (Low and High), race (White/Asian and Black), and gender (Female and Male). The Low-SES and non-White distributions lie to the left of their counterparts across essentially the entire distribution, with median differences of 0.15 and 0.09 standard deviations, respectively, and the largest gap occurs in the lower half, where the least familiar candidates are concentrated. For gender, the two curves are indistinguishable, consistent with the absence of a gap in the regressions. At the same time, the figure shows substantial variation in familiarity within every group. Unfamiliar candidates exist in all groups, even if they are more common in some than in others. These two facts are complementary. The shift in means is what makes the test biased: it penalizes some groups at any ability level, and averaging over candidates does not remove it. Within-group variation shows that familiarity is ultimately an individual characteristic, and group labels capture it only partially. This is why we measure familiarity at the individual level and evaluate interventions that reach every unfamiliar candidate, while the burden of the status quo still falls disproportionately on Low-SES and non-White test takers.

6.2 Familiarity's Effect on Performance

To test whether familiarity affects performance, we use the experimental variation in participants' familiarity. All participants read an article on a given topic. Some individuals then write the essay on the same topic as the article, and others on a different one. The article should increase individuals' familiarity with a given topic, and then we can assess whether that increase translated into better performance. We do this in three steps. First, we test whether reading an article on a given topic increased familiarity with that topic. Then, we test whether reading an article on the same topic as the essay task increased writing performance. Finally, through an IV strategy, we measure how much of the change in performance is due to familiarity.

Table 5: Familiarity's Effect on Performance

	First Stage	Reduced Form	IV
Dependent Variable	Familiarity (std)	Essay Score (std)	Essay Score (std)
Treated	0.121***	0.134**	
	(0.046)	(0.065)	
Familiarity (std)			1.112**
			[0.201, 4.937]
FE: post, order, topic, treated, id	Yes		
FE: essay topic		Yes	
Controls		Yes	
Observations	2260	579	579

Note: This table shows the results of three regressions. The first column represents the first stage: treatment effect on familiarity with the respective fixed effects. The second column is the reduced form: treatment effect on performance, with the respective fixed effect and control variables. For these two regressions, we show individual-level clustered standard errors in parentheses. $^{***}: 0.01, ^{**}: 0.05, ^{*}: 0.1$. The third column shows the IV estimate, and in brackets, we show the Anderson-Rubin 95% confidence interval, and the significance level is based on the Anderson-Rubin test. All these regressions compare groups T and C_1.

Table 5 shows the results. First, reading a text on a given topic increases familiarity by 0.12 s.d. A small intervention of about 3 to 5 minutes moves the median participant to the 54th percentile of familiarity. Note that this effect also corresponds to the size of the SES familiarity gap described in the previous section, and is one and a half times the racial gap.

Next, we test whether reading an article on the same topic as the essay directly affects the score. We find that individuals who read about the essay topic before the task perform 0.13 s.d. better than individuals who read about a different one. The effect corresponds to 0.14 points on the 9-band IELTS scale[13] and between 0.8 and 1 point on the TOEFL 0-30 scale.[14] The median essay in our control group is scored at band 6.0, in line with the average IELTS Academic Writing performance of test takers worldwide. The treatment effect moves the median participant from the 50th to the 62nd percentile of the control-group score distribution. If we benchmark against the worldwide TOEFL Writing distribution, an improvement of this size moves a test taker of that level from the 39th to roughly the 48th percentile of all test takers.

Interestingly, the performance improvement is uniform across the four evaluation

[13] IELTS reports scores in half-band increments; 0.14 points is about a quarter of one increment.

[14] Using the ETS concordance between TOEFL iBT and IELTS Writing scores (ETS linking study), which maps one IELTS band to 6 to 7 TOEFL writing points in the range of our sample.

criteria: lexical resource, grammar, task response, and coherence and cohesion. This is an important finding, particularly regarding the visible increases in lexical resource and grammar, which measure language ability rather than content-related knowledge (Figure A6 in the Appendix). This is consistent with the mechanism proposed by schema theory (Bartlett, 1932), which argues that writers with an activated schema for the topic do not need to divert attention and working memory to generating and organizing content, and can allocate those resources to the language dimensions of the task (Robinson and Gilabert, 2007). This shows that increasing familiarity has important spillovers, since if we expected it to affect only the essay's content, we should observe improvements only in task response and the coherence and cohesion categories. Even if one believes these exams should partly evaluate individuals' knowledge of the chosen topics, they must conclude that familiarity with the topic also distorts measures that should assess language ability alone.

One concern is that this result comes from the control group losing confidence in writing an essay about a topic they previously declared unfamiliar, which would then hinder their performance. We address this concern in Section 8 and show no evidence that asking about familiarity before the essay led to lower scores.

Finally, we estimate the effect of familiarity on scores using an IV approach. The IV's exclusion restriction is that the article affects essay scores only through what participants know and feel about the topic. Our design supports this: the article is written in Portuguese, so it cannot improve English proficiency. Moreover, because control participants also read an article, only on a different topic, any effect of reading itself, such as engagement, warm-up before writing, or fatigue, is common to both groups and cannot drive the estimate. Therefore, the exclusion restriction can fail only through topic-specific channels other than familiarity, such as direct translation or content transfer. Since participants have no access to the article while writing, this reduces these concerns. In fact, in the robustness tests in Section 8, we show that the existing overlap between the article and the essay reflects on-topic knowledge, not rewarded transfer, and does not explain the score effect.

We find that a one-standard-deviation increase in familiarity increases the essay score by 1.1 s.d. We obtain a wide 95% confidence interval of [0.20, 4.94], as expected given the moderate first stage, but significant at the 5 percent level.[15] Note that

[15]We use the Anderson-Rubin test for inference because the estimator is a ratio whose sampling

its lower bound of 0.2 s.d. is itself a meaningful effect, four times the 0.05 s.d. estimate that Lee and Schaelling (2026) obtain from observational variation in content relatability, and about three times the 0.07 gradient we document in Section 6.1. The gap between the correlational estimates and the IV is expected, since self-reported familiarity is measured with noise, biasing correlational gradients downward and the IV upward.[16]

Combined with the familiarity disparities documented in the previous section, we can estimate $\beta \times \gamma$ from Equation (3). This estimate implies that topic-familiarity bias alone accounts for between 0.12 and 0.14 standard deviations of the High-SES score advantage, over 50% of the gap we observe in the control group.[17] Even at the lower bound of the confidence interval, familiarity is responsible for about 10% of the SES performance gap. Regarding the racial gap, familiarity can account for about 0.09 s.d. of the score difference, which is comparable in size to the entire racial gap we observe in the control group.[18]

6.2.1 Heterogeneous Effects

The previous section established that familiarity matters for performance. The article raised participants' familiarity with the essay topic, and this increase translated into higher writing scores. This finding is relevant for test designers, as it highlights a source of construct-irrelevant variance in test scores. Moreover, because familiarity with test topics differs systematically across social groups, this design feature does not affect all test takers equally. It places an additional burden on groups that are already disadvantaged, such as Low-SES students. Exploring design alternatives that reduce this bias is therefore an important step toward fairer tests.

In this section, we test whether our intervention, giving an article on the essay topic for the test-taker to read in their native language, is successful in closing the

distribution does not follow a normal distribution.

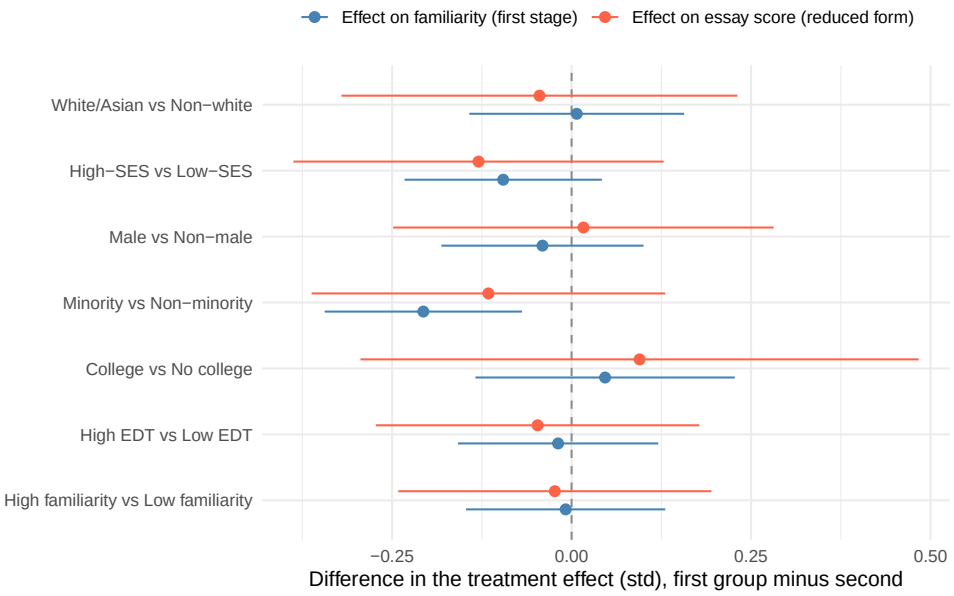
[16] We obtain very similar results by excluding f_1 from the familiarity measure. Results are presented in the Appendix A.2.

[17] The range reflects whether we use the SES familiarity gap measured over all 30 topics (0.13 s.d.) or over the 6 experimental topics (0.11 s.d.).

[18] This comparison, however, should be taken with caution because the control-group racial score gap is imprecisely estimated (0.09 standard deviations, p = 0.35). We therefore read this decomposition as indicating that familiarity disparities are large enough to matter for racial score differences, but not as a precise accounting.

documented gaps. To do so, we look at treatment effects by race, socioeconomic status, gender, minority status, college education, score on the English Diagnostic Test (EDT), and baseline familiarity. The estimates are noisy because sample sizes are small within each characteristic, but overall we do not find that our intervention differentially changes individuals' familiarity by social group. The only exception is that the treatment seems more effective at increasing familiarity among individuals who self-declare as non-minorities and, marginally, among those with Low-SES (Figure 2). The same pattern holds for performance. We find no evidence that the treatment led to different performance gains across groups. Even for self-declared non-minority individuals, for whom we saw a larger increase in familiarity, this did not translate into statistically significant performance gains. If anything, their performance increased less, but this effect is not statistically different from zero. (Figure 2).

Figure 2: Heterogeneous Treatment Effects on Familiarity and Performance



Note: Each point shows the difference in the treatment effect between the two groups of a dimension (first-listed minus second), estimated separately for the first stage (familiarity) and the reduced form (essay score). Specifications follow Table 5, with the treatment indicator interacted with group membership. Bars show 95% confidence intervals based on individual-level clustered standard errors. The sample includes the graded, human-written essays of groups T and C_1.

Taken together, these results suggest that providing topic information raises familiarity and performance across the board rather than favoring any particular group. An intervention of this kind can therefore lift the performance of unfamiliar test takers

wherever they are found, but it does not eliminate the group-level gaps documented in Section 6.1 on its own, since treated participants from more familiar groups also benefit.

7 Alternative Intervention

With the article intervention, we test whether manipulating f_{ik} directly affects scores, holding the assigned topic k fixed. However, we can also address the documented bias in a second way. An alternative design lets each candidate select the topic k they are most comfortable with. This effectively allows test takers to maximize their own familiarity f_{ik}, closing their familiarity deficit through selection rather than information.

To assess whether letting test-takers choose their preferred topic closes the familiarity gap, we randomly allocated some participants to an alternative intervention. In Part I, these participants were asked about their familiarity with the six experimental topics, plus the two anchors. In Part II, they did not read any article; instead, they chose, from those six topics, which they wanted to write about in their essay task. They could also be assigned a randomly selected topic.

This design also strongly validates our familiarity measure, since we can test the relationship between baseline familiarity with each topic and participants' choices. First, 83% of the participants chose their most familiar topic. Then we find that the chosen topic is 0.4 standard deviations more familiar than the non-chosen topics (See Table A6). This matters because a survey answer given days earlier strongly predicts a subsequent decision, which is incentivized since the 10% best essays receive a large bonus.

We then test whether the alternative intervention (A) led to higher scores than the article intervention (T), compared to all control groups (C). Table 6 shows the results of this analysis. In column (2), we see that treatment A increases the essay score by 0.13 s.d. more than treatment T. This difference is underpowered, so it is not statistically significant, but the point estimate is economically meaningful. In fact, when we compare both interventions to all control groups, we find a significant advantage of A over T of 0.14 s.d., significant at 10%. We also confirm that both interventions increased writing performance, with A increasing scores by 0.27 s.d. over the control group.

Note that participants in the alternative intervention score 0.27 s.d. above the control group and write on topics that are 0.4 s.d. more familiar than their alternatives

(column (3) of Table A6). This implies a score-to-familiarity ratio of roughly 0.65, within the Anderson-Rubin interval of our IV estimate. Topic choice is not a clean manipulation of familiarity, so we read this only as a consistency check, but it is reassuring that a second, independent source of familiarity variation points to an effect of the same order.

Table 6: Comparing Treatments T and A

Dependent Variable:	Essay Score (std)			
Model:	(1)	(2)	(3)	(4)
Variables				
Free Choice (A)	0.1490	0.1321	0.2946***	0.2666***
	(0.0982)	(0.0823)	(0.0961)	(0.0802)
Treated (T)			0.1480**	0.1260**
			(0.0753)	(0.0618)
Observations	419	419	778	778
FE: essay topic		Yes		Yes
Controls		Yes		Yes
Sample	T vs A	T vs A	T vs A vs C	T vs A vs C

Note: This table compares the topic-choice intervention (A) with the article intervention (T) and the control groups. The dependent variable is the essay score (std). In columns (1) and (2), the sample includes groups T and A, so the coefficient on A is the difference between choosing and reading. In columns (3) and (4), both interventions are compared with the pooled control groups. Columns (2) and (4) add essay-topic fixed effects and controls.Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

Allowing individuals to write about a topic of their choice is the intervention that leads to higher essay scores. However, does it work differently across social groups? In Table 7 we show the heterogeneous effects of both treatments across race, socioeconomic status, gender, self-declared minority status, education, and baseline English ability. Qualitatively, the alternative intervention increases the racial gap, particularly compared with T, but decreases the SES gap more than T. It also reduces the minority gap, since it benefits more self-declared non-minorities, who reported lower familiarity levels than self-declared minority individuals. The article seems to benefit participants with completed college more, whereas choosing benefits those who haven't completed yet. These comparisons, however, need to be taken cautiously. The magnitudes of the differences are meaningful, but none of them are statistically significant.

Table 7: Heterogeneous Treatment Effects on Performance, by Intervention

	Article (T)	Free choice (A)	p-value: T = A
White/Asian vs Non-white	-0.007	0.105	0.472
	(0.136)	(0.161)	
High-SES vs Low-SES	-0.090	-0.174	0.590
	(0.125)	(0.155)	
Male vs Non-male	-0.010	0.005	0.921
	(0.130)	(0.160)	
Minority vs Non-minority	-0.115	-0.261*	0.352
	(0.121)	(0.154)	
College vs No college	0.129	-0.117	0.287
	(0.191)	(0.232)	
High EDT vs Low EDT	0.017	0.065	0.751
	(0.108)	(0.140)	
High fam vs Low fam	0.046	0.072	0.883
	(0.105)	(0.154)	

Note: Each cell shows the difference in the intervention's effect on the essay score (std) between the two groups of a row (first group minus second), estimated in one regression per row with both interventions against the pooled control groups. The last column reports the p-value of the test that the group difference is the same for the two interventions. Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

There are three conclusions we can take from this analysis. First, both interventions improve performance relative to the status quo of assigning a topic with no information: providing an article increases scores by 0.13 standard deviations, and allowing topic choice increases scores by 0.26 standard deviations, with the difference favoring choice. Second, we do not find that either intervention benefits some groups significantly more than others: the group differences in treatment effects are imprecisely estimated, and none is statistically distinguishable from zero. Our experiment is powered to detect average effects, but not interactions, so these results should be read as ruling out large differential effects rather than establishing equal ones, given that the point estimates are often economically meaningful. Third, the absence of detectable differential effects does not mean the interventions are irrelevant for fairness. Both operate by raising familiarity, which, as we showed, causally affects performance and is unequally distributed across groups. An intervention that increases familiarity of whoever lacks it, whether by providing information or by letting candidates avoid unfamiliar topics, therefore shrinks familiarity biases in test scores, even if we cannot measure group-specific effects precisely at this sample size. On practical grounds, topic choice seems more attractive for closing the gap for Low-SES students, although

it might widen the racial gap. The article intervention also seems effective at shrinking the SES gap, but not the racial one.

Note that the two interventions are complements. The first, the article, is a clean manipulation because it moves only familiarity, and it does so exogenously. This is what allows us to interpret the IV as a causal estimate. It is also an intervention used in real-world standardized exams, so a rigorous evaluation of it is directly policy relevant. The second, the topic choice alternative, is a bundled intervention. It raises familiarity but may also work through motivation, past practice, or other dimensions we do not observe. Our design cannot separate these mechanisms. However, its policy relevance does not depend on this decomposition. Since 83% of candidates choose their most familiar topic, choice reduces the influence of topic luck on scores, whatever the combination of channels behind the average gain.

8 Robustness

As briefly discussed in the Results Section, there are three potential threats to the validity of our results that should be discussed.

Familiarity Measure Our familiarity measure is based on self-reported survey answers. A natural concern is that these do not reflect real background knowledge or experiences. Participants could overstate or understate their familiarity, or their answers may be confounded with other personality traits. The anchoring procedure described in Section 4.1 addresses differences in scale usage, but it cannot address the deeper concern that the answers carry no meaningful information. In this section, we present three exercises showing that the measure captures something real that we relate highly to familiarity and background knowledge.

First, as described in Section 7, familiarity predicts incentivized decisions. In the alternative intervention arm, participants overwhelmingly chose the topic they had rated as most familiar days earlier. This shows that familiarity translates into confidence in writing (See Table A6).

Second, familiarity correlates positively with general knowledge. In the baseline survey, we have asked individuals general knowledge questions on a subset of topics for which they also reported their familiarity. We observe that the larger the reported

familiarity, the larger is the share of correct answers in the quiz (see Table A9 in the Appendix).

Third, our familiarity measure predicts performance. Recall that all participants in the control group answered their familiarity level on two of the six experimental topics. Then, they were randomly assigned one of those topics to write the essay about, but were given no information on it. Among those, some were 'lucky', in the sense that they were assigned to the topic they were most familiar with at baseline, compared to the ones that were assigned to the least familiar one. We find that the lucky individuals had 0.18 s.d. higher scores than the unlucky ones, statistically significant at 5% (see the score distribution for the two groups in Figure A7 in the Appendix). Note that, following the framework of Section 2, this exercise estimates the effect of a random draw of f_{ik}. Because topic assignment is random, the comparison is independent of group membership and would exist even if $\gamma = 0$. In addition to showing that our measure of familiarity predicts performance, it also quantifies the topic-luck channel that the framework distinguishes from statistical and systematic bias.

A final concern is that our familiarity measure captures more than familiarity, for example, ability. High-ability candidates may have read more, discussed more, and feel more capable of arguing about any topic. If our measure was just capturing that, then the familiarity gaps of Section 6.1 would just be ability gaps. To address this concern, we re-estimate the familiarity gaps controlling for the two ability measures collected in Part I, the English diagnostic test and the critical thinking scale.[19] Table A10 shows that the gaps are only slightly smaller. Conditional on measured ability, High-SES participants are 0.10 s.d. more familiar with the test topics than Low-SES participants, about three-quarters of the unconditional gap, and White and Asian participants are 0.05 s.d. more familiar, about two-thirds. Familiarity does correlate with ability, but most of the gap is not explained by the latter. Candidates with the same English score and the same critical thinking score still differ systematically in how familiar they are with the topics of the test.

Together, these exercises show that our measure predicts choices, knowledge, and

[19] We do not include ability controls in the specifications of Section 6.1. The familiarity gap entering the decomposition is the unconditional gap between groups, since that gap generates the bias. Moreover, ability is itself correlated with group membership and partly produced by the same background that generates familiarity, so conditioning on it absorbs part of the gap we aim to measure. The conditional estimates in this section are therefore a conservative lower bound, informative about the measure rather than about the size of the bias.

performance, thus constituting a meaningful measure of familiarity.

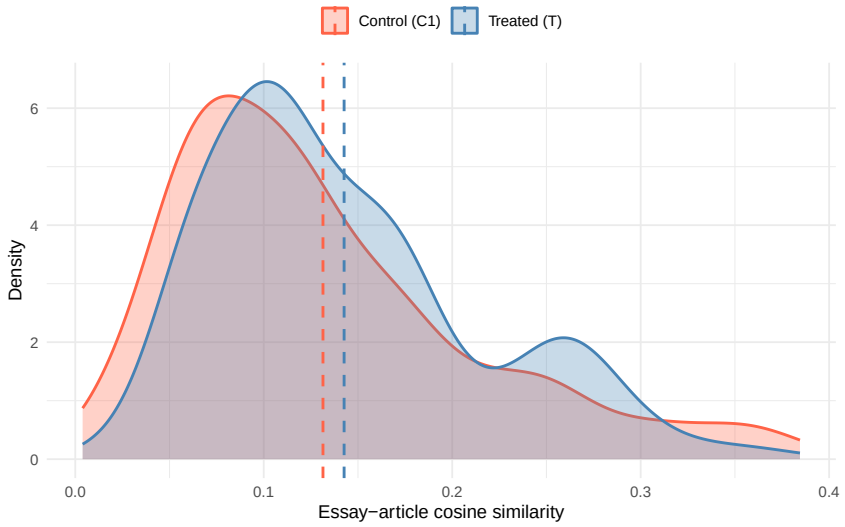
Exclusion Restriction Another worry is that the article not only increases knowledge and familiarity, but also improves writing performance by directly providing information that can be translated into the essay. Two reasons make this unlikely: participants do not have access to the text while writing the essay, and the article does not provide direct answers to the essay prompt. Nevertheless, we conduct a text analysis to compare the content of the articles with the essays. Note that some similarity between the article and the essay is expected, given that they are on the same topic. Indeed, among control participants who never saw an article on the same topic as their essay, the mean cosine similarity is 0.12, compared to 0.03 for an unrelated article. Participants who read the article on the same topic as their essay have higher similarity than those who did not, as shown in Figure 3. Word overlap could also miss transfer expressed in different words, so we repeat the exercise with a semantic measure, comparing each essay to the original Portuguese article using multilingual sentence embeddings. By this measure, treated essays are no closer to the article than control essays. The difference is 0.008 on a control-group mean of 0.59 and is statistically insignificant (column (2) of Table A7).

Similarity also does not explain the effect on scores. Columns (3) to (5) of Table A7 regress scores on similarity among control participants only, who never saw the relevant article. Similarity strongly predicts their scores, indicating that it mainly reflects on-topic content rather than transferred information from the article. When we consider the two measures simultaneously, lexical similarity carries all the predictive power. Moreover, if scores reflected translated material, similarity should predict scores more strongly among treated participants. Table A8 shows the opposite: the interaction between similarity and treatment is negative and statistically insignificant. Finally, the similarity channel is too small to account for the treatment effect. Multiplying the treated increase in similarity by the control-group gradient implies a contribution of at most 0.06 s.d., well below the 0.13 s.d. effect in this sample, and even this is conservative, since similarity is itself an outcome of increased familiarity.

Finally, we also observe a positive correlation between reading the article and knowledge obtained. Individuals answered questions on the topic before (in Part I) and after reading the article. Reading the article is associated with a 7.7 p.p. increase in the share

of correct answers (Table A5 in the Appendix). Overall, the evidence indicates that the article works by changing what participants know about the topic rather than by giving content to reproduce.

Figure 3: Cosine Similarity between Article and Essay, by Treatment Group



Note: The figure shows the distributions of essay-article cosine similarity by treatment group.

Discouragement of the Control Group The third concern is whether reminding an individual of their unfamiliarity with a topic before writing about it has negative effects. To assess whether this is a valid concern, we compare the performance of the control group $C1$ with that of the two alternative control groups, $C2$ and $C3$. Recall that the latter groups are in everything similar to $C1$ until the essay task, except that they do not answer familiarity questions until that point. Reassuringly, we find no statistically significant differences in scores across groups. This result holds if we restrict the sample to individuals with low baseline familiarity with the essay. These estimates are noisier due to the sample size, but if anything the magnitude would go against the hypothesis.

Table 8: The Effect of Asking for Familiarity

Dependent Variable:	Essay Score (std)					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Constant	0.0406			-0.1852		
	(0.0979)			(0.1433)		
C1 (asked familiarity before essay)	-0.0406	-0.0788	-0.0367	0.0723	0.0497	0.0547
	(0.1144)	(0.1104)	(0.0892)	(0.1688)	(0.1609)	(0.1344)
Observations	359	359	357	176	176	173
FE: essay topic		Yes	Yes		Yes	Yes
Controls			Yes			Yes
Sample	All	All	All	Low fam.	Low fam.	Low fam.

Note: This table tests whether asking about familiarity immediately before the essay depresses scores. Group C_1 answered familiarity questions in Part II before writing, while groups C_2 and C_3 did not. The dependent variable is the essay score (std), and the sample includes the three control groups. Columns (4) to (6) restrict the sample to participants with low baseline familiarity with their essay topic. Columns (2), (3), (5), and (6) add essay-topic fixed effects, and columns (3) and (6) also add controls.Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

9 Conclusion

Every year, millions of people take standardized tests whose scores determine access to education, jobs, and migration. Often, tests measure the skills of interest within contexts or topics that are not part of the evaluation per se. This paper shows that this contextualization matters. Through a large online randomized experiment, we show that familiarity with the topics used in standardized English tests affects writing performance. Furthermore, we show that familiarity with the test topic is unequally distributed across social groups, making it a source of test bias and unfairness. We estimate that the socioeconomic familiarity gap accounts for over half of the performance gap observed between High-SES and Low-SES test takers.

This gap has concrete consequences. In our control group, the median High-SES participant writes at band 6.0 on the IELTS scale, while the median Low-SES participant writes at band 5.5. The most common writing cutoff for graduate admission, band 6.0, falls exactly between the two: the median High-SES candidate meets it and the median Low-SES candidate does not. Graduate programs typically require an overall IELTS score between 6.5 and 7.5, and most also impose a minimum score on the writing component, usually between 6.0 and 7.0. The same thresholds matter for

migration. Australia's skilled migration system awards additional points at writing bands 7.0 and 8.0, and Canada's Express Entry requires band 6.0 in writing for eligibility. Closing the SES familiarity gap alone would raise Low-SES scores by 0.12 to 0.14 band points, and letting candidates choose their topic raises scores by 0.27 band points, half of the distance between the median Low-SES candidate and the threshold.

Should testing organizations act on this? The two interventions we evaluate increase performance at almost no cost. Providing a short informational text or letting candidates choose among topics lets test takers focus on how to write rather than what to write about. The second is not part of what the test evaluates, but as we show, it spills over into the first. Whether these designs also narrow group gaps, our experiment can answer only imprecisely: we find no evidence of differential gains across groups, with suggestive evidence that choice narrows the socioeconomic gap but not the racial one.

A test that asks candidates to write about a museum is, for those who have never visited one, partly a test of the visit. Test designers cannot make every topic equally familiar to everyone, but they can measure the gaps, supply the missing background, or let candidates write about what they know.

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A Appendix

A.1 Tables and Figures

Table A1: Topics

#	Topic
1	**Some people think the money spent on developing technology for space exploration is not justified. They are more beneficial ways to spend this money.**
2	**Team activities can teach more skills for life than other activities which are played alone.**
3	**Children can learn effectively through watching TV. Therefore, children should be encouraged to watch TV regularly both at home and at school.**
4	**It is better to have your own small business than it is to work for a salary at a large company.**
5	**Some people believe that the best way to preserve the environment and slow down climate change is to stop eating meat.**
6	**The best way to learn a foreign language is to speak it with native speakers.**
7	More attention is paid to being "responsible tourists" in order to preserve the cultural and environmental aspects of tourist places. However, some people think it is impossible to be a "responsible tourist" in the society.
8	It is not necessary to travel to other places to learn about other cultures of other people. We can learn just as much from books, films, and the internet.
9	Museums and art galleries should focus on works that show the history and culture of their own country rather than works of other parts of the world.
10	In most countries, prison is the common solution for the problem of crime. However, some think the more effective solution is to provide people with better education so that people will not become criminals.

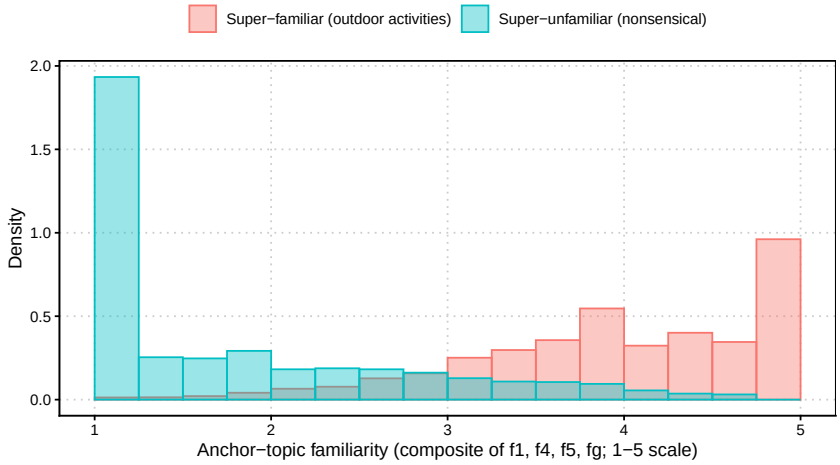
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#	Topic
11	In the past, the role of teacher was to provide information. Today, students have access to wide sources of information. There is, therefore, no need for the role of teacher in modern education.
12	Nowadays in many countries women have full time jobs. Therefore, it is logical to share household tasks evenly between men and women.
13	Today people are surrounded by advertising. This affects what people think is important and has a negative impact on people's lives.
14	Some countries spend a lot of money preparing competitors to take part in major competitions such as Olympic Games or football World Cup. Some people say that it would be better to spend this money encouraging children to take up sports from a young age.
15	The restoration of old buildings in major cities throughout the world often involves numerous government expenditure. Some people think this money will bring more benefits if it is used to provide new housing or road development.
16	The best way for governments to solve the problem of traffic congestion is providing free public transport in 24 hours per day, and seven days a week.
17	Some people think the best way to solve the environmental problems is to increase the cost of fuel for cars and other vehicles.
18	Traffic and housing problems in major cities could be solved by moving large companies and factories and their employees to the countryside.
19	In the future, bicycles will be replaced by a more convenient means of transportation.
20	Children who use electronic devices earn better grades than children who do not.
21	Dancing plays an important role in a culture.
22	Every individual should spend at least a year living in a foreign country.

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#	Topic
23	In the Industrial Age, many developed countries aimed for more significant economic growth every generation. While they are making considerable progress, the setbacks cannot be eliminated. Economic development shares a close relationship with environmental protection, and in the long run, the latter reflects more importance.
24	It is better to read books on an electronic device than it is to read traditional books on paper.
25	It is important that young people need to learn some domestic skills, like cooking, sewing, and taking care of children.
26	Parents are the best teachers.
27	Primary schools should spend more time teaching young students (5–11 years old) technology (like computers) than teaching them music and art.
28	Professional athletes are born with talent. You cannot train someone to be a professional athlete.
29	Teenagers should work while they go to school so they can learn how to be more responsible.
30	The government should add taxes to unhealthy foods to improve people's health.

Figure A1: Familiarity with the Anchoring Topics



Note: The figure shows the distribution of familiarity with the two anchoring topics, computed as the average of the four items used in the main measure (f_1, f_4, f_5, f_g) on the original 1 to 5 scale. The super-familiar anchor states that outdoor activities are more enjoyable when it is not raining. The super-unfamiliar anchor is a nonsensical statement about the dynamic exploration of extra-marginal resources. The sample includes participants satisfying quality criteria I.

Table A2: Final Sample: Treatment Allocation

Treatment Arm	N	Share	N	Share	Retention
A	603	0.151	127	0.163	0.211
C1	1,473	0.370	287	0.369	0.195
C2	201	0.050	33	0.042	0.164
C3	200	0.050	39	0.050	0.195
T	1,505	0.378	292	0.375	0.194
Total	3,982	1.000	778	1.000	0.195
Attrition test (p-value)					0.684

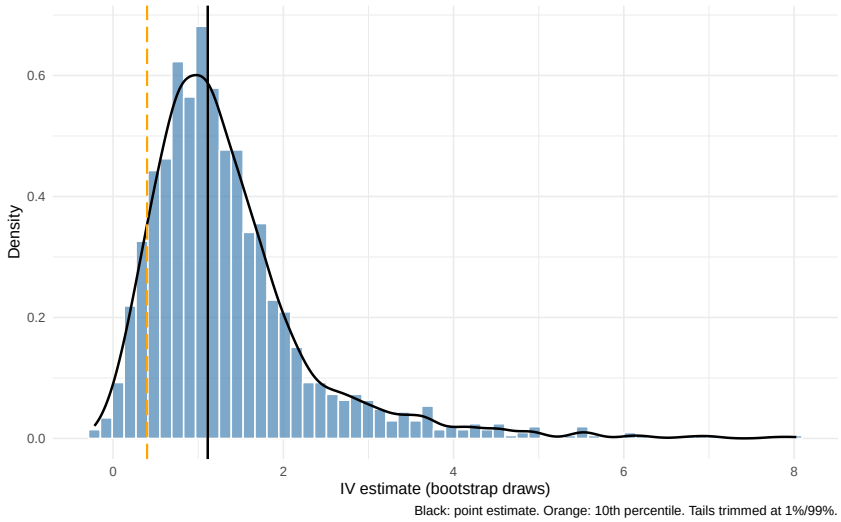
Note: This table shows the proportion of individuals in each treatment arm. The sample includes individuals who satisfied Criteria I and II and have a human-written essay.

Table A3: Final Sample: Balance Test

	All	T	C1	C2	C3	A	p-value
High-SES	0.534	0.562	0.532	0.576	0.526	0.465	0.461
	(0.499)	(0.497)	(0.500)	(0.502)	(0.506)	(0.501)	
Male	0.357	0.372	0.332	0.364	0.256	0.409	0.335
	(0.480)	(0.484)	(0.472)	(0.489)	(0.442)	(0.494)	
White/Asian	0.656	0.647	0.676	0.636	0.711	0.621	0.761
	(0.475)	(0.479)	(0.469)	(0.489)	(0.460)	(0.487)	
Self-Minority	0.488	0.455	0.477	0.545	0.487	0.571	0.253
	(0.500)	(0.499)	(0.500)	(0.506)	(0.506)	(0.497)	
Age	25.728	25.514	25.784	26.818	26.385	25.606	0.678
	(5.685)	(5.772)	(5.794)	(6.478)	(4.446)	(5.382)	
College Degree or above	0.851	0.860	0.836	0.848	0.872	0.858	0.934
	(0.356)	(0.348)	(0.371)	(0.364)	(0.339)	(0.350)	
English Diagnostic Test	8.513	8.586	8.495	8.515	8.769	8.307	0.815
	(2.442)	(2.320)	(2.479)	(2.774)	(2.389)	(2.574)	
Critical Thinking Scale	46.186	45.990	46.146	46.152	46.590	46.614	0.681
	(4.239)	(4.111)	(4.241)	(5.075)	(4.089)	(4.364)	
N	778	292	287	33	39	127	

Note: This table shows the balance of the demographics and control variables used in the regressions, across treatment arms. The sample includes individuals who satisfied Criteria I and II and have a human-written essay. The p-value is from a joint equality-of-means test across the five treatment arms, computed separately for each row variable.

Figure A2: Bootstrap Distribution of the IV Estimates



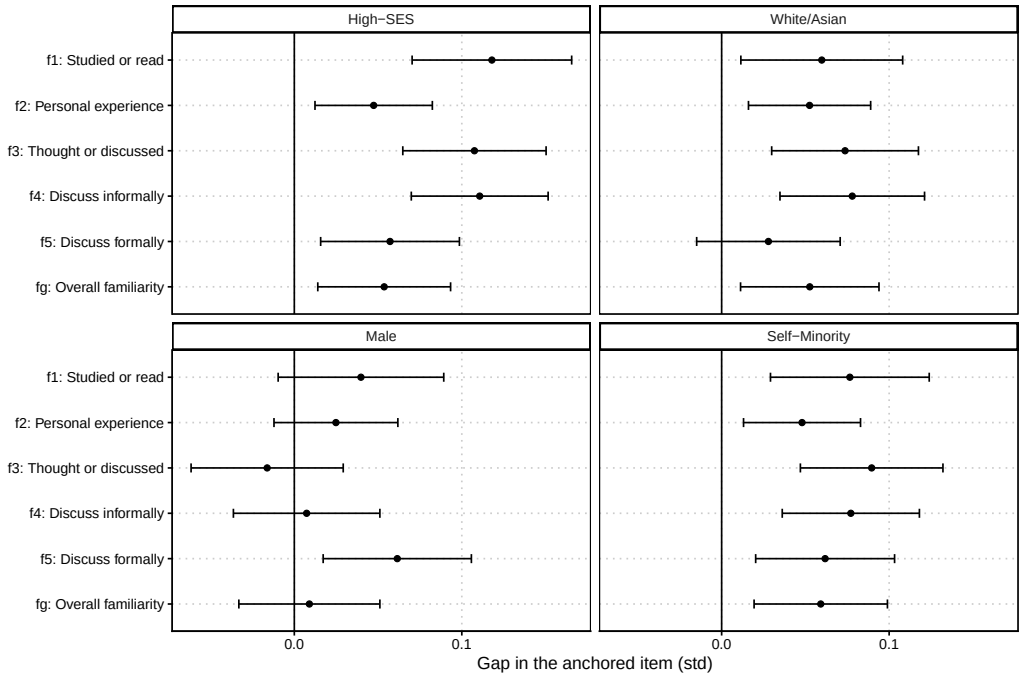
Note: The figure shows the distribution of the IV estimator across 1,500 Bayesian bootstrap draws. Each draw assigns Dirichlet weights at the individual level and re-estimates both stages jointly, preserving their correlation. The first stage uses our main familiarity measure. The solid black line marks the point estimate (1.11); the dashed orange line marks the 10th percentile of the draws. For readability, we do not show draws below the 1st and above the 99th percentile.

Table A4: AI Use by Treatment Arm

Dependent Variable:	AI-written Essay			
Model:	(1)	(2)	(3)	(4)
Variables				
Treated (T)	-0.0043	0.0006	-5.24×10^{-5}	0.0078
	(0.0286)	(0.0291)	(0.0270)	(0.0275)
Free Choice (A)			0.0416	0.0309
			(0.0367)	(0.0358)
Observations	701	701	951	951
FE: essay topic		Yes		
Controls		Yes		Yes
Sample	T, C1	T, C1	All arms	All arms

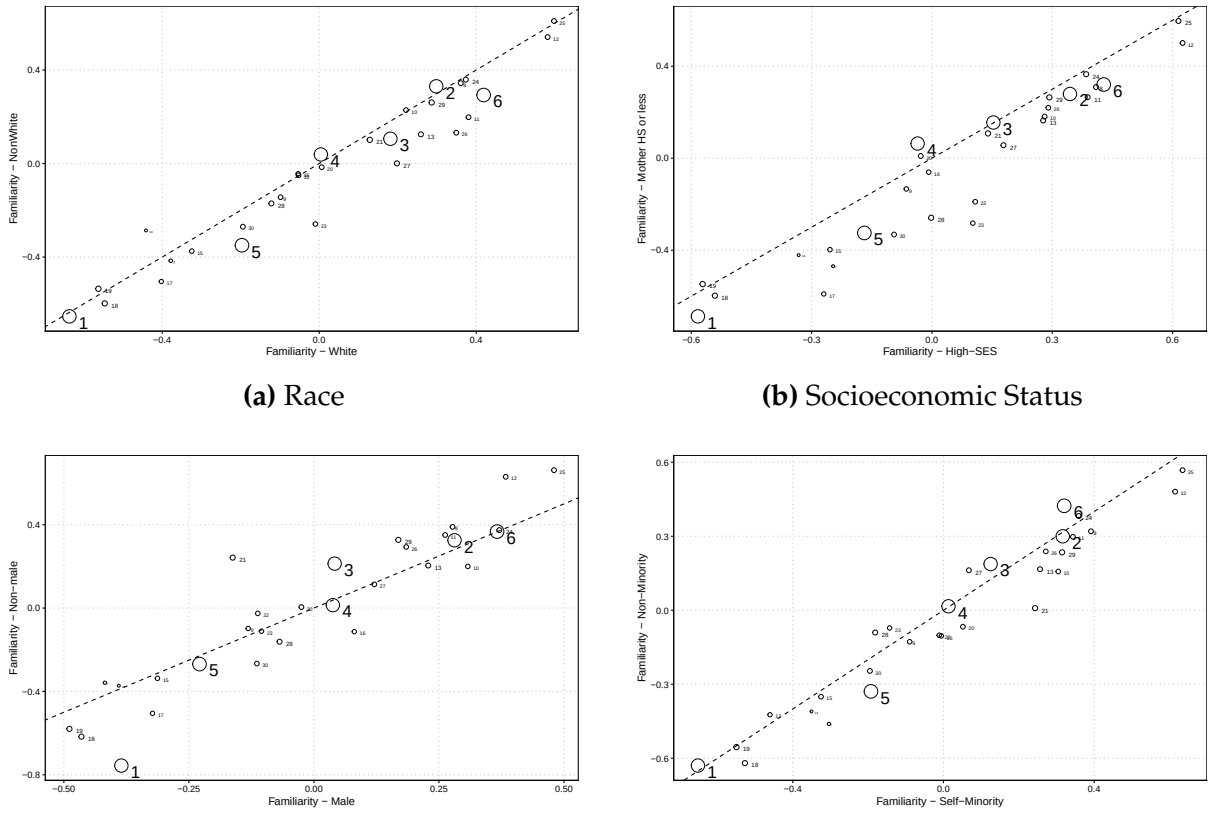
Note: This table shows that treatment assignment does not predict AI use. The dependent variable is an indicator for the essay being flagged by *Pangram* as not fully human-written. Columns (1) and (2) compare groups T and C_1, and columns (3) and (4) include all arms, with an indicator for each treatment. Columns (2) and (4) add controls, and column (2) essay-topic fixed effects. The sample includes all graded essays satisfying criteria I and II, before excluding AI-written essays. Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

Figure A3: Familiarity Gaps by Item of the Familiarity Measure



Note: The figure reports the group coefficients from regressing each item of the familiarity measure on the group indicators. Panels show the coefficients for High-SES, White/Asian, Male, and self-declared minority, respectively. Regressions include age, the male indicator, and all group indicators jointly, with topic and question-order fixed effects, following the specification of Table 4. The sample includes participants satisfying quality criteria I, with one observation per participant and topic for the 30 topics, measured before treatment. Standard errors are clustered at the participant level and bars show 95% confidence intervals.

Figure A4: Average Familiarity by Topic, across Groups

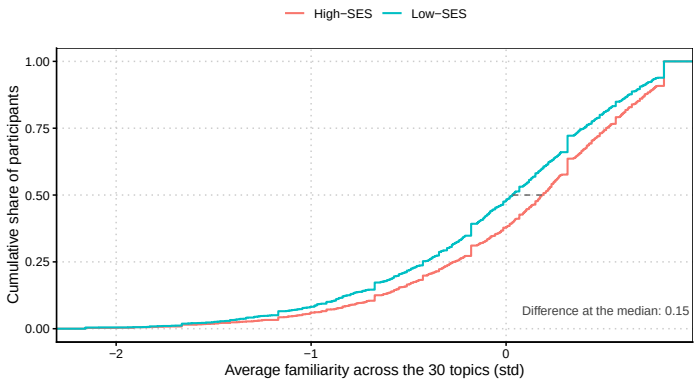


(a) Race **(b)** Socioeconomic Status

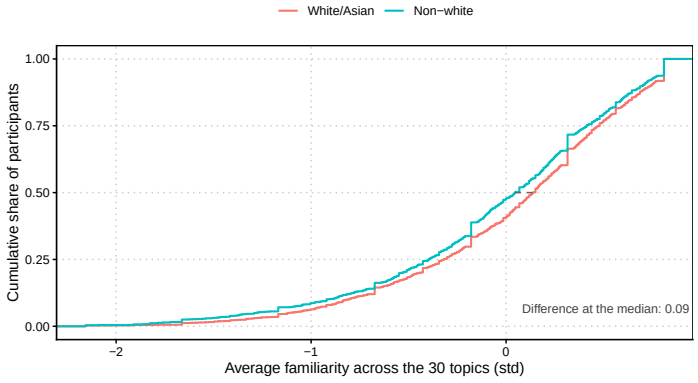
(c) Gender **(d)** Self-perceived Minority

Note: Each point is one of the 30 test topics, with point size proportional to the number of respondents rating the topic. Axes show the average anchored familiarity measure (standardized across all 30 topics) for each group, and the dashed line is the 45-degree line. Points below the line are topics more familiar to the group on the horizontal axis. The sample includes participants satisfying quality criteria I.

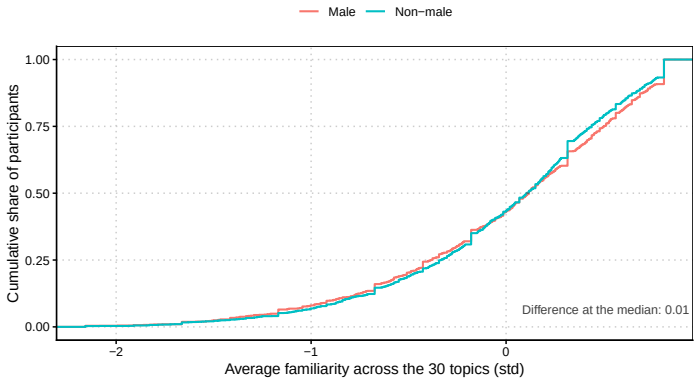
Figure A5: Distribution of Familiarity within Groups



(a) SES



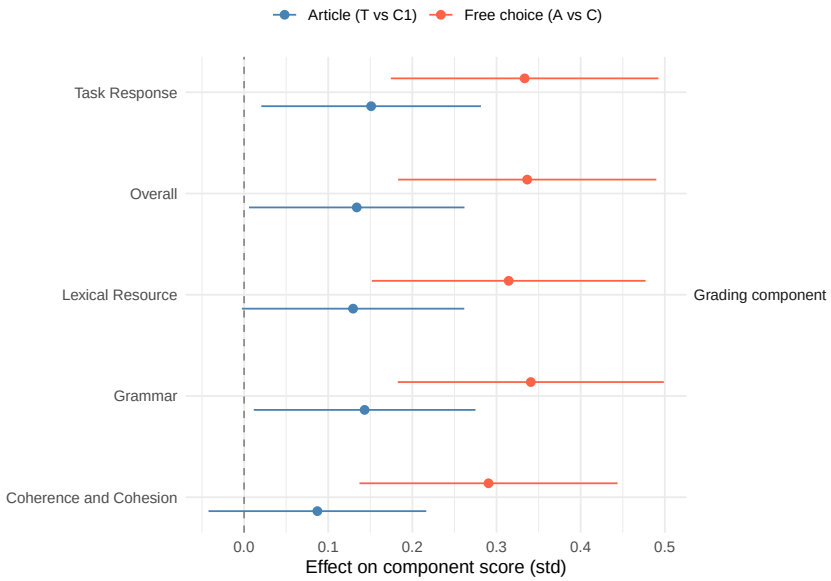
(b) Race



(c) Gender

Note: Each panel plots the empirical cumulative distribution of familiarity, with one observation per participant, defined as the average familiarity across the 30 topics (anchoring-adjusted measure, standardized). The sample includes participants satisfying quality criteria I with non-missing group information. Panel (a) compares High-SES and Low-SES participants, panel (b) White and Asian participants against all others, and panel (c) male against non-male participants. The dashed segment marks the horizontal distance between the two group medians, and the text in each panel reports this difference.

Figure A6: Treatment Effects on Scores, by Evaluation Criteria



Note: The figure shows the treatment effect on the overall essay score and on each of the four evaluation criteria. Each outcome is standardized using the control-group mean and standard deviation. Specifications follow the reduced form of Table 5. Bars show 95% confidence intervals based on individual-level clustered standard errors. The sample includes the graded, human-written essays of groups T and C_1.

Table A5: Treatment Effect on Knowledge

Dependent Variable:	Knowledge (share correct)		
Model:	(1)	(2)	(3)
Variables			
Constant	0.7782***		
	(0.0090)		
Post (after reading)	0.0769***	0.0769***	0.0769***
	(0.0090)	(0.0090)	(0.0090)
Observations	1,304	1,304	1,304
FE: topic		Yes	Yes
FE: id			Yes

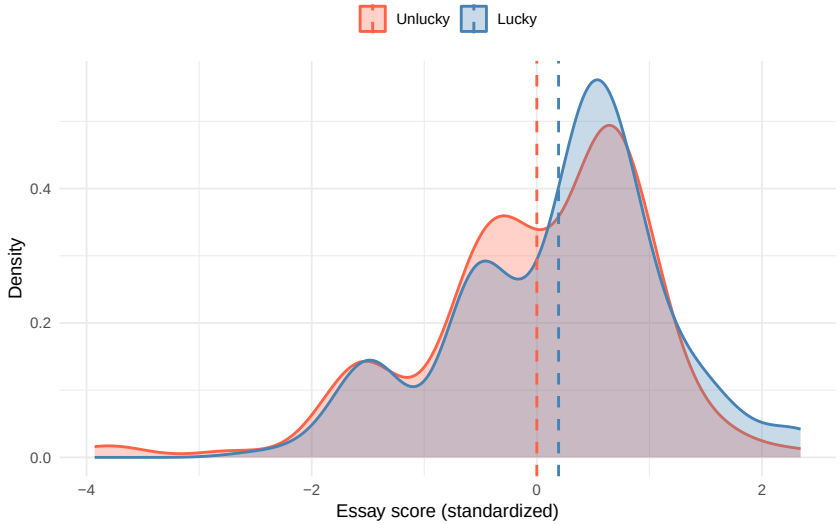
Note: This table shows the effect of reading the article on knowledge about its topic. The dependent variable is the share of correct answers on the general knowledge questions about the read topic, asked before (Part I) and after (Part II) reading the article. Post is an indicator for the post-reading measurement. Column (2) adds topic fixed effects and column (3) individual fixed effects, so the estimate compares the same individual before and after reading. The sample includes participants in the final sample who read an article.Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

Table A6: Familiarity of the Chosen Topic

Dependent Variable:	Familiarity (std)		
Model:	(1)	(2)	(3)
Variables			
Constant	0.0472		
	(0.0507)		
Chosen topic	0.5714***	0.5733***	0.4157***
	(0.0504)	(0.0506)	(0.0526)
Observations	737	737	737
FE: id		Yes	Yes
FE: topic			Yes

Note: This table shows the familiarity premium of the chosen topic among participants in the alternative intervention (arm A). The dependent variable is the Survey I familiarity (std) with each of the six menu topics. Chosen topic is an indicator for the topic the participant later chose for the essay. Column (2) adds individual fixed effects. Column (3) adds topic fixed effects, so the estimate reflects the within-person familiarity premium of the chosen topic net of generally popular topics. Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

Figure A7: Score Distribution for Lucky and Unlucky Individuals



Note: This figure shows the score distribution of individuals in the control groups depending on the topic they were randomly assigned to write. The control group answers baseline familiarity with two of the six experimental topics. Individuals who are lucky are the ones who wrote the essay on their most familiar topic, and those who are unlucky wrote the essay on their least familiar topic.

Table A7: Text Similarity between Essay and Article

Dependent Variables:	Lexical Similarity	Semantic Similarity	Essay Score (std)		
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Treated	0.0109***	0.0082			
	(0.0040)	(0.0072)			
Lexical Similarity			5.594***		5.119***
			(1.200)		(1.123)
Semantic Similarity				1.752*	0.9030
				(0.9548)	(0.9061)
Observations	579	579	287	287	287
FE: essay topic	Yes	Yes	Yes	Yes	Yes
R^2	0.57688	0.56403	0.10685	0.05769	0.11224
Within R^2	0.01313	0.00223	0.07402	0.02305	0.07960
Sample	T, C1	T, C1	C1	C1	C1
Control group mean (cosine)	0.121	0.589			

Note: Lexical similarity is the cosine similarity between TF-IDF-weighted word vectors of the essay and of the English translation of the article on the essay topic. Semantic similarity is the cosine similarity between multilingual sentence embeddings of the essay and of the original Portuguese article (model paraphrase-multilingual-mpnet-base-v2). Columns (1) and (2) regress each similarity measure on the treatment indicator in the sample of graded, human-written essays of the T and C1 groups, with the control-group means reported for reference. Columns (3) to (5) restrict the sample to control participants, who never saw an article on their essay topic, and regress scores on lexical similarity, on semantic similarity, and on both jointly. All regressions include essay-topic fixed effects. Standard errors are clustered at the participant level. *** : 0.01,** : 0.05,* : 0.1.

Table A8: Semantic Essay-Article Similarity and Scores

Dependent Variable:	Essay Score (std)		
Model:	(1)	(2)	(3)
Variables			
Treated	0.1339**	0.2492*	0.3274
	(0.0653)	(0.1270)	(0.3141)
Lexical Similarity		3.315***	
		(0.7697)	
Lexical Similarity × Treated		-1.163	
		(0.8289)	
Semantic Similarity			0.4746
			(0.5396)
Semantic Similarity × Treated			-0.3307
			(0.5111)
Observations	579	579	579
FE: essay topic	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Sample	T, C1	T, C1	T, C1

Note: Column (1) reports the treatment effect on the essay score in this sample. Columns (2) and (3) interact the treatment indicator with the lexical and the semantic essay-article similarity measures of Table A7. The similarity coefficients give the score gradient among control participants, who never saw the relevant article. The sample includes the graded, human-written essays of the T and C_1 groups. All regressions include essay-topic fixed effects. Individual-level clustered standard errors in parentheses. $^{***}: 0.01, ^{**}: 0.05, ^{*}: 0.1$.

Table A9: Correlation between Familiarity and General Knowledge

Dependent Variable:	Knowledge (share correct)	
Model:	(1)	(2)
Variables		
Constant	0.6396***	
	(0.0090)	
Familiarity (std)	0.0170**	0.0215**
	(0.0075)	(0.0093)
Observations	2,021	2,021
FE: topic, id		Yes

Note: Each cell shows the difference in the intervention's effect on the essay score (std) between the two groups of a row (first group minus second), estimated in one regression per row with both interventions against the pooled control groups. The last column reports the p-value of the test that the group difference is the same for the two interventions. Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

Table A10: Familiarity Gaps with and without Ability Controls

Dependent Variable:	Familiarity (std)			
Model:	(1)	(2)	(3)	(4)
Variables				
Age	-0.0013	-0.0011	0.0030	0.0037
	(0.0017)	(0.0017)	(0.0023)	(0.0023)
Male	0.0210	0.0134	0.0377	0.0329
	(0.0216)	(0.0214)	(0.0295)	(0.0294)
White/Asian	0.0735***	0.0470**	0.0815***	0.0503*
	(0.0210)	(0.0210)	(0.0285)	(0.0286)
High-SES	0.1293***	0.0982***	0.1100***	0.0734***
	(0.0198)	(0.0199)	(0.0268)	(0.0270)
Self-Minority	0.0720***	0.0619***	0.0307	0.0241
	(0.0201)	(0.0199)	(0.0274)	(0.0271)
EDT (std)		0.0811***		0.1020***
		(0.0117)		(0.0157)
CTS (std)		0.0658***		0.0473***
		(0.0115)		(0.0150)
Observations	22,602	22,602	9,817	9,817
FE: topic	Yes	Yes	Yes	Yes
Topics	All 30	All 30	6 experimental	6 experimental
Ability controls	No	Yes	No	Yes

Note: This table compares the familiarity gaps with and without ability controls. All columns regress familiarity with the topic (anchoring-adjusted, standardized) on all group indicators jointly, with age, the male indicator, and topic fixed effects, as in column (5) of Table 4. Columns (1) and (2) use the 30 test topics, and columns (3) and (4) the six experimental topics. Even columns add the standardized English diagnostic test and critical thinking scale scores as controls, with indicators for missing scores. One observation per participant and topic, measured before treatment. The sample includes participants satisfying quality criteria I. Individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1.

A.2 Results with pre-registered Familiarity Measure

In the pre-analysis plan, we pre-registered a familiarity measure based on f_4, f_5, and f_g, which excludes the knowledge component f_1. Table A11 replicates our main results using this pre-specified measure. The reduced form is identical by construction, since it does not involve the familiarity measure. The first stage is smaller, 0.09 s.d. (against 0.12 s.d.), and remains statistically significant. This is expected: the article operates partly by increasing knowledge about the topic, which is precisely the dimension the pre-specified measure leaves out, so this measure captures only part of the change in familiarity that the treatment induces. The IV estimate is correspondingly larger (1.45), since dividing the same reduced form by a smaller first stage mechanically raises the

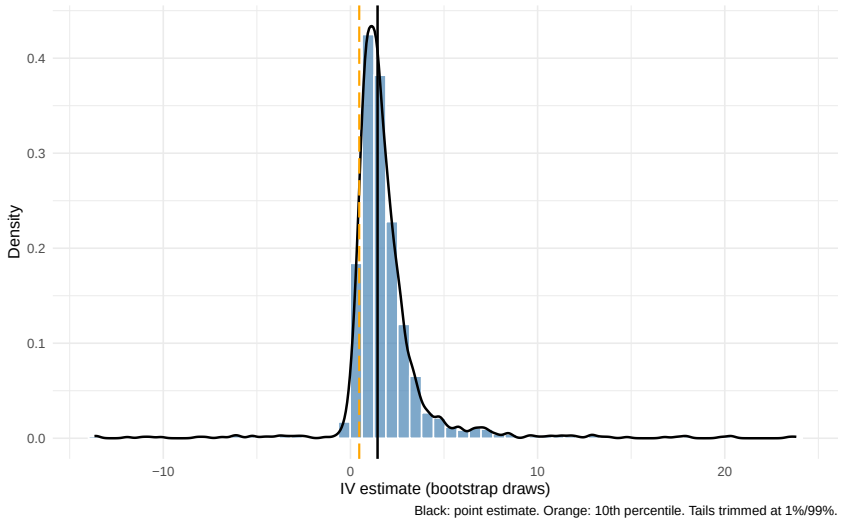
ratio, and its Anderson-Rubin interval is wider, since a weaker first stage translates into less informative inference. The interval still excludes zero. All qualitative conclusions are unchanged. Figure A8 shows the bootstrap distribution of the IV estimator under this measure, which displays the same right-skewed shape as the main measure and motivates the same Anderson-Rubin inference.

Table A11: Familiarity's Effect on Performance: Pre-specified Familiarity Measure

	First Stage	Reduced Form	IV
Dependent Variable	Familiarity (std)	Essay Score (std)	Essay Score (std)
Treated	0.093**	0.134**	
	(0.047)	(0.065)	
Familiarity (std)			1.449**
			[0.269, 52.539]
FE: post, order, topic, treated, id	Yes		
FE: essay topic		Yes	
Controls		Yes	
Observations	2316	579	579

Note: This table replicates Table 5 using the pre-specified familiarity measure based on f_4, f_5, and f_g (excluding f_1). The first column shows the first stage, the second column the reduced form, and the third column the IV estimate, with the Anderson-Rubin 95% confidence interval in brackets, and the significance level is based on the Anderson-Rubin test. For the first two columns, individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1. All regressions compare groups T and C_1.

Figure A8: Bootstrap Distribution of the IV Estimates: Pre-specified Familiarity Measure



Note: The figure shows the distribution of the IV estimator across 1,500 Bayesian bootstrap draws, with the first stage based on the pre-specified familiarity measure (f_4, f_5, f_g). Each draw assigns Dirichlet weights at the individual level and re-estimates both stages jointly, preserving their correlation. The solid black line marks the point estimate; the dashed orange line marks the 10th percentile of the draws. For readability, we do not show draws below the 1st and above the 99th percentile.

A.3 AI-written Essays

Our main sample excludes the 166 essays that *Pangram* flags as not fully human-written. Table A4 shows that treatment assignment does not predict AI use, so this exclusion does not introduce differential selection across the arms used in the main analysis. Table A12 reports the main results when these essays are kept in the sample, which adds 121 graded essays in arms T and C_1 (700 instead of 579). The first stage is essentially unchanged (0.11 against 0.12 s.d.), as familiarity questions are answered by the participant regardless of how the essay was produced. The reduced form falls from 0.13 to 0.10 s.d. and is no longer statistically significant. This attenuation is almost exactly what mechanical dilution predicts: the treatment cannot affect the quality of an essay written by a language model, and rescaling the human-only estimate by the share of human essays in the pooled sample ($0.134 \times 579/700 = 0.111$) accounts for nearly all of the drop. The IV estimate falls accordingly, and its Anderson-Rubin interval no longer excludes zero, as expected given the diluted reduced form. We therefore read this exercise as confirming that AI-written essays add noise that is unrelated to treatment, rather than as evidence against the effect: the human-written sample is the

relevant one for studying test-taker behavior, and the pattern of dilution is what genuinely machine-written essays should produce.

Table A12: Familiarity's Effect on Performance: Including AI-written Essays

	First Stage	Reduced Form	IV
Dependent Variable	Familiarity (std)	Essay Score (std)	Essay Score (std)
Treated	0.112***	0.101	
	(0.043)	(0.066)	
Familiarity (std)			0.897
			[-0.211, 4.359]
FE: post, order, topic, treated, id	Yes		
FE: essay topic		Yes	
Controls		Yes	
Observations	2736	700	700

Note: This table replicates Table 5 keeping the essays flagged as AI-written in the sample. The first column shows the first stage, the second column the reduced form, and the third column the IV estimate, with the Anderson-Rubin 95% confidence interval in brackets. For the first two columns, individual-level clustered standard errors in parentheses. *** : 0.01,** : 0.05,* : 0.1. All regressions compare groups T and C_1. Essay scores are standardized using the control-group mean and standard deviation of this larger sample.